演化生物学生信算法

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王克桢楼306

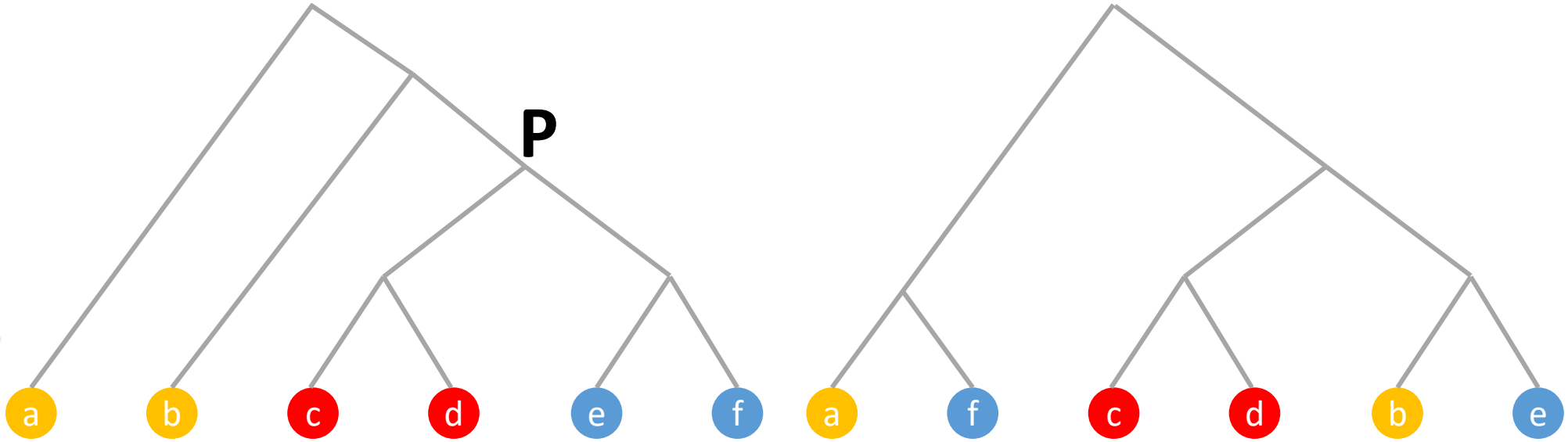
北京大学 PEKING UNIVERSITY

生命科学联合中心 CENTER FOR LIFE SCIENCES



内容回顾

公共四叶树问题

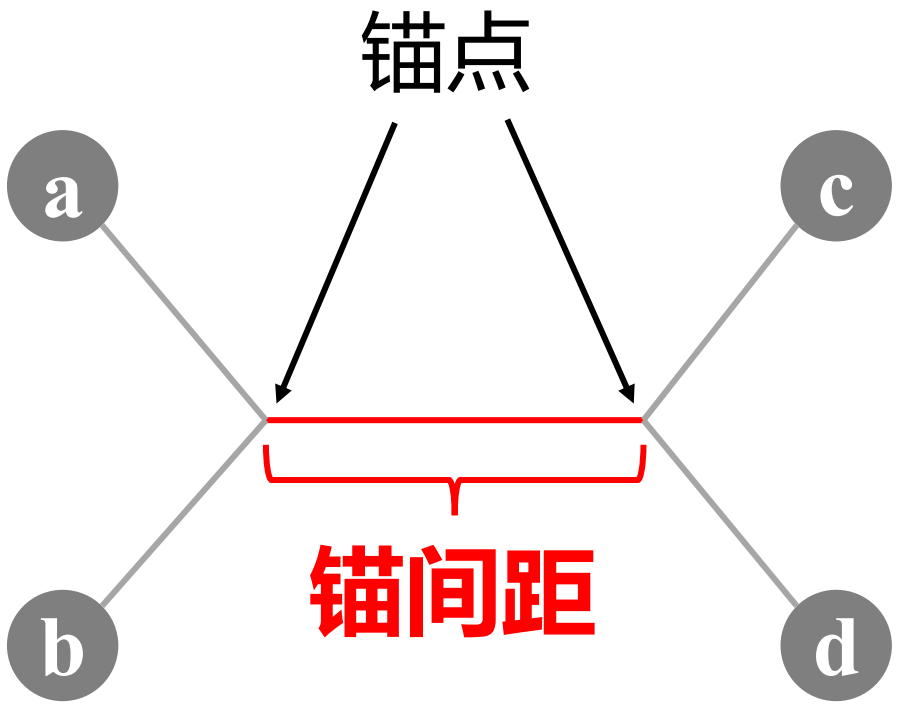


遍历左树所有中间节点P – 共$n-1$个

- 对P左子树的**叶子节点**染红色、右子树的**叶子节点**染蓝色、其他**叶子节点**染金色 – $O(n)$
- 计算右树中**三色四叶树**的数量 – $O(n)$

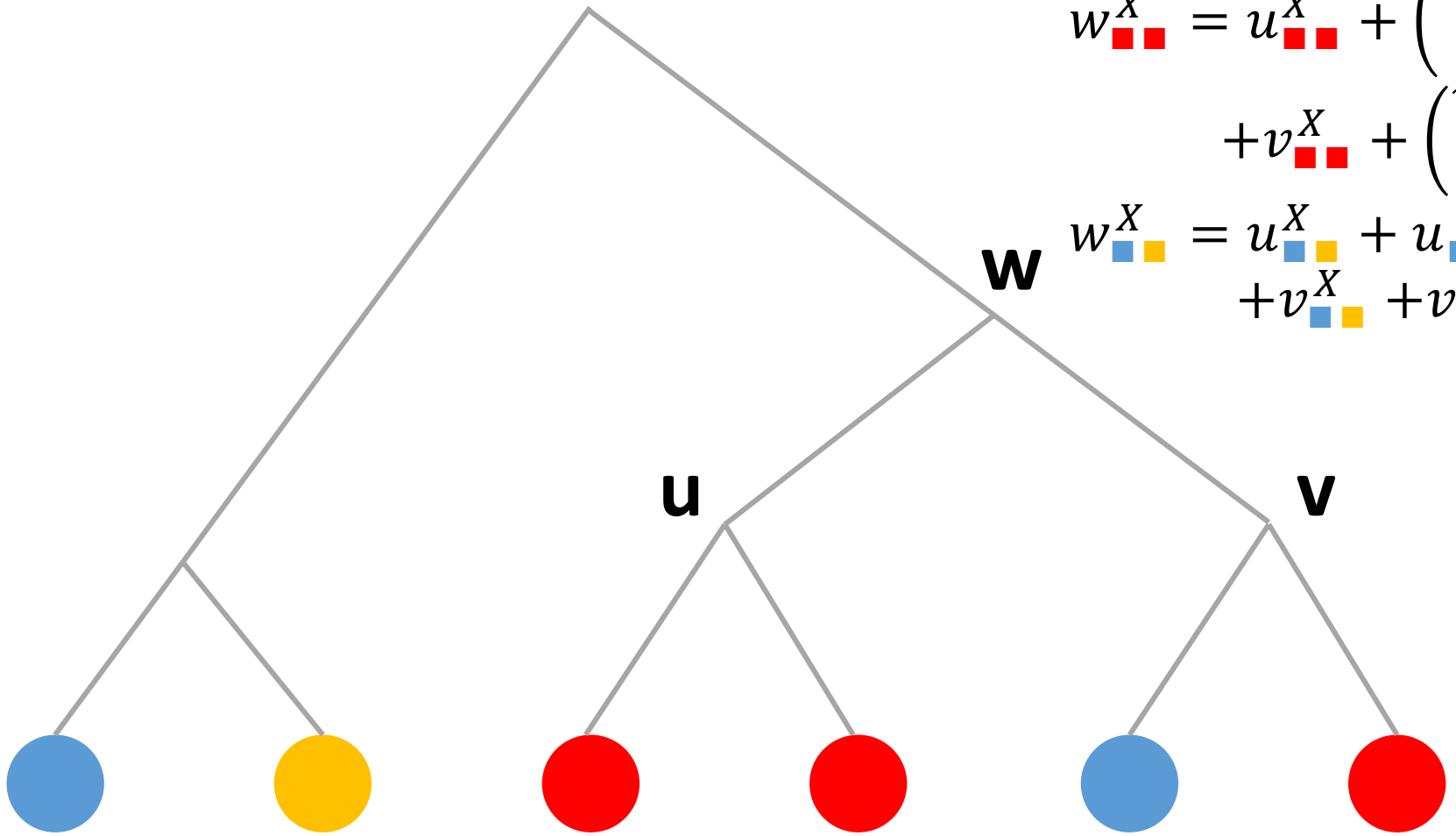
公共四叶树数量即三色四叶树的数量之和除以2?

定义：（无根）四叶树锚间距



四叶树两个**锚点**间的距离称为**锚间距**。

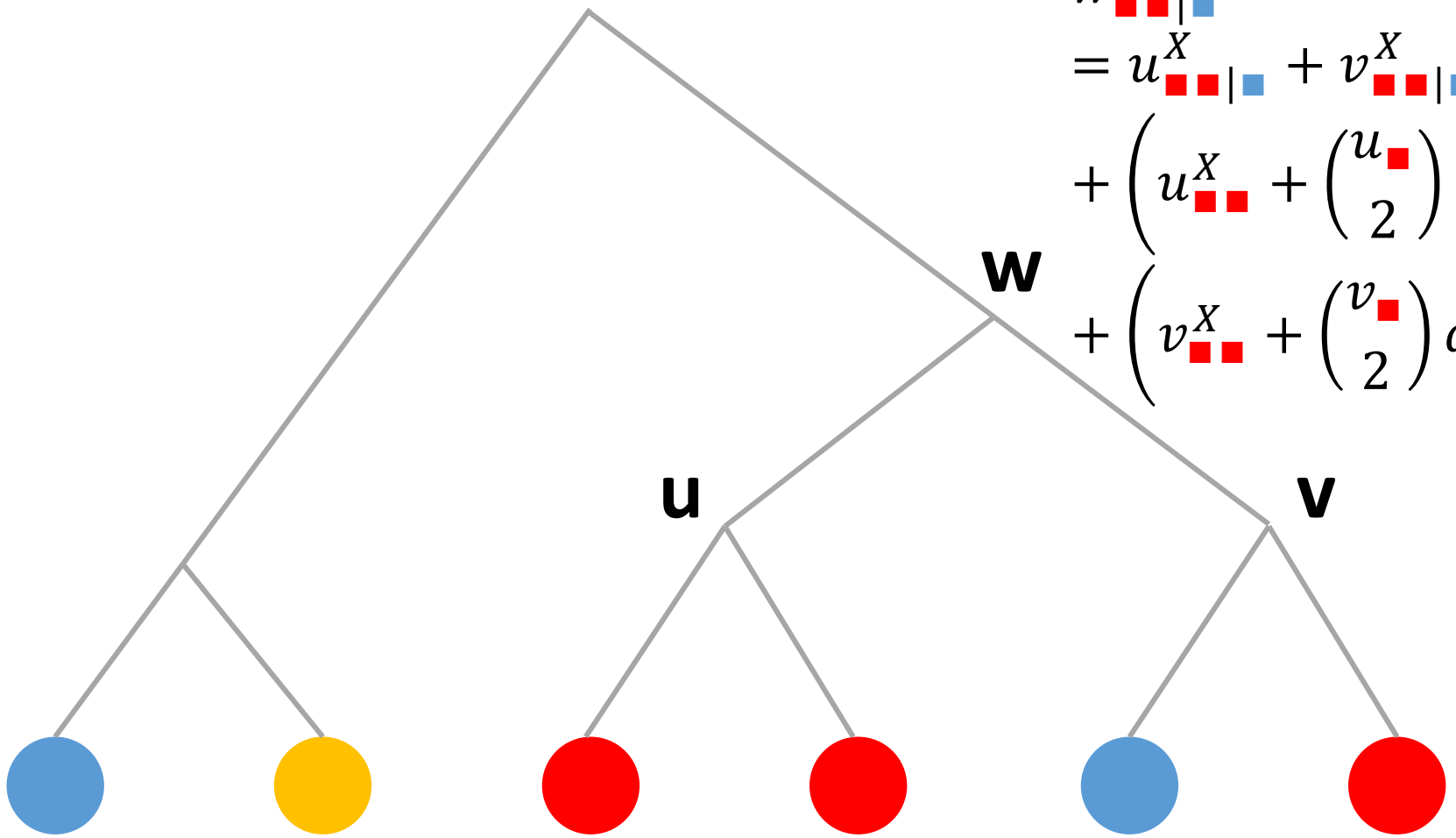
红红蓝金四叶树总锚间距问题



$$w_{\blacksquare\blacksquare}^{X} = u_{\blacksquare\blacksquare}^{X} + \binom{u_{\blacksquare}}{2} d(w,u) + v_{\blacksquare\blacksquare}^{X} + \binom{v_{\blacksquare}}{2} d(w,v)$$

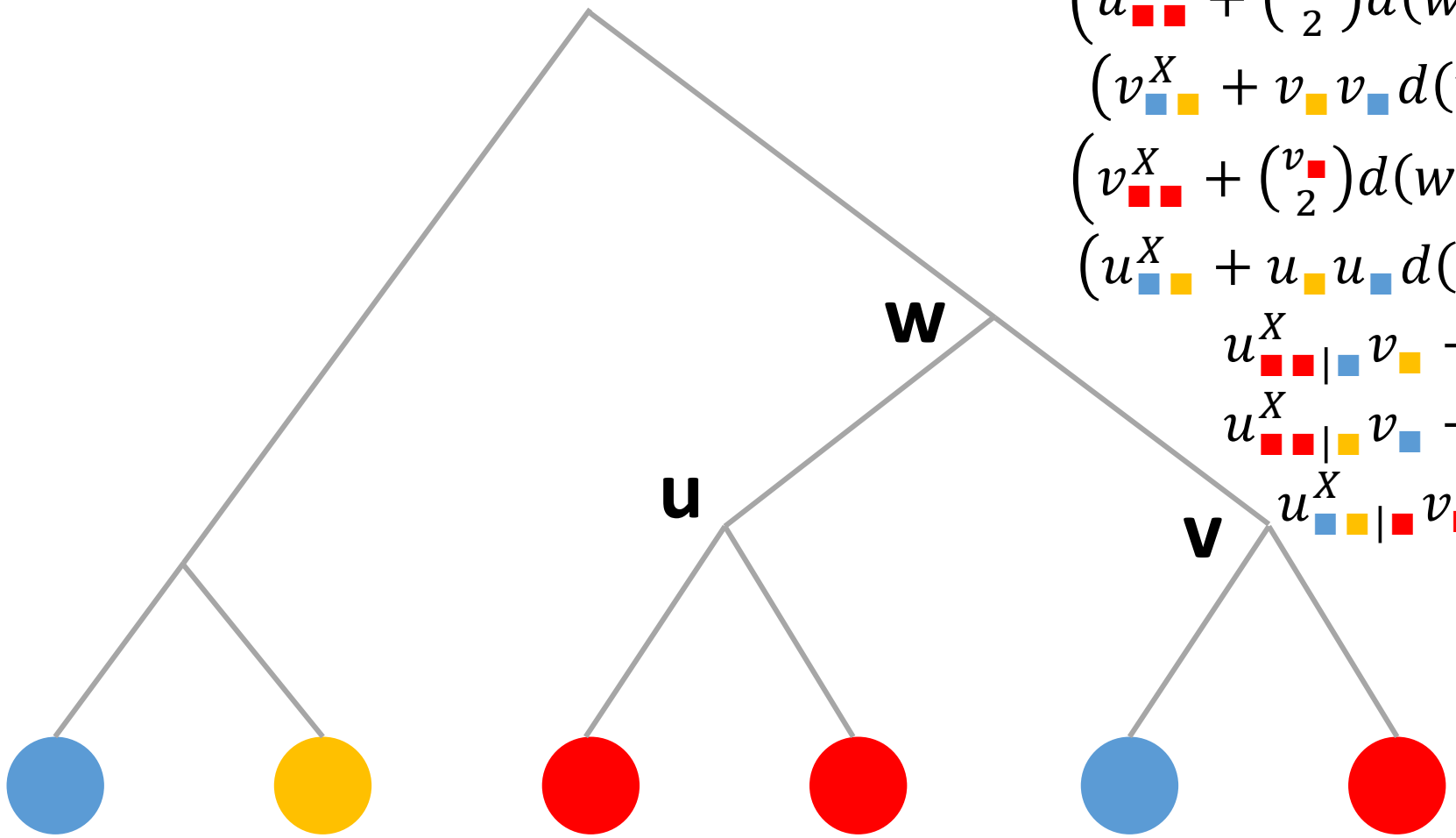
$$w_{\blacksquare\blacksquare}^{X} = u_{\blacksquare\blacksquare}^{X} + u_{\blacksquare} u_{\blacksquare} d(w,u) + v_{\blacksquare\blacksquare}^{X} + v_{\blacksquare} v_{\blacksquare} d(w,v)$$

红红蓝金四叶树总锚间距问题



$$w^X_{\blacksquare\blacksquare|\blacksquare} = u^X_{\blacksquare\blacksquare|\blacksquare} + v^X_{\blacksquare\blacksquare|\blacksquare} + \left(u^X_{\blacksquare\blacksquare} + \binom{u_{\blacksquare}}{2} d(w,u)\right) v_{\blacksquare} + \left(v^X_{\blacksquare\blacksquare} + \binom{v_{\blacksquare}}{2} d(w,v)\right) u_{\blacksquare}$$

红红蓝金四叶树总锚间距问题

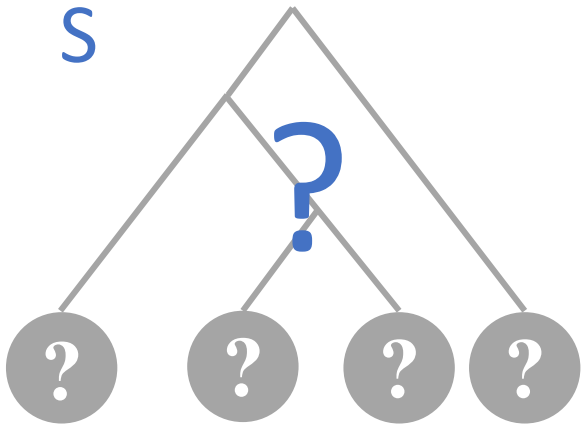


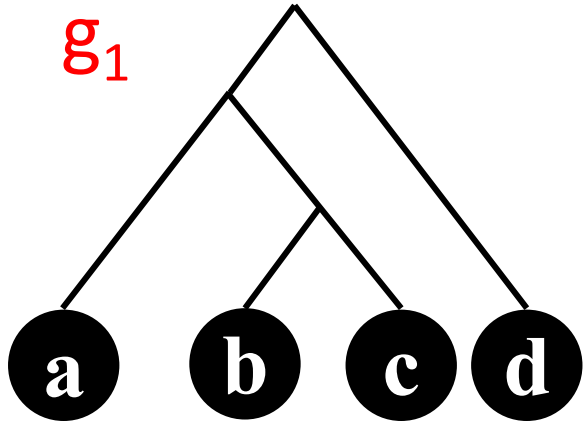
$$\left(u^{X}_{\blacksquare\blacksquare} + \binom{u_{\blacksquare}}{2} d(w,u)\right) v_{\blacksquare} v_{\blacksquare} +$$
$$\left(v^{X}_{\blacksquare\blacksquare} + v_{\blacksquare} v_{\blacksquare} d(w,v)\right) \binom{u_{\blacksquare}}{2} +$$
$$\left(v^{X}_{\blacksquare\blacksquare} + \binom{v_{\blacksquare}}{2} d(w,v)\right) u_{\blacksquare} u_{\blacksquare} +$$
$$\left(u^{X}_{\blacksquare\blacksquare} + u_{\blacksquare} u_{\blacksquare} d(w,u)\right) \binom{v_{\blacksquare}}{2} +$$
$$u^{X}_{\blacksquare\blacksquare|\blacksquare} v_{\blacksquare} + v^{X}_{\blacksquare\blacksquare|\blacksquare} u_{\blacksquare} +$$
$$u^{X}_{\blacksquare\blacksquare|\blacksquare} v_{\blacksquare} + v^{X}_{\blacksquare\blacksquare|\blacksquare} u_{\blacksquare} +$$
$$u^{X}_{\blacksquare\blacksquare|\blacksquare} v_{\blacksquare} + v^{X}_{\blacksquare\blacksquare|\blacksquare} u_{\blacksquare}$$

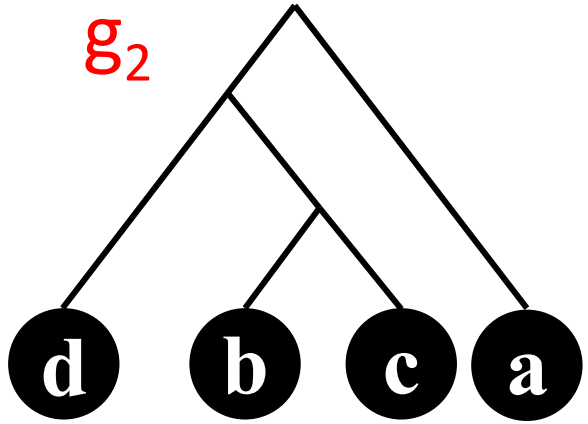


12.三叶最优树问题

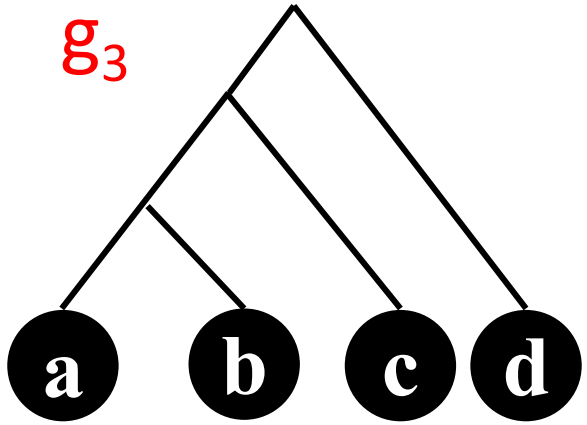
12.1 三叶最优树问题

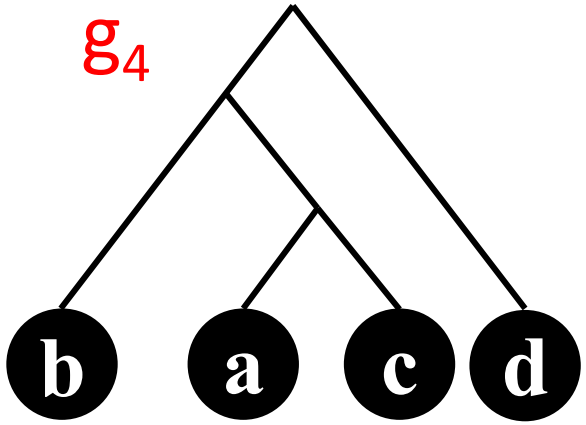




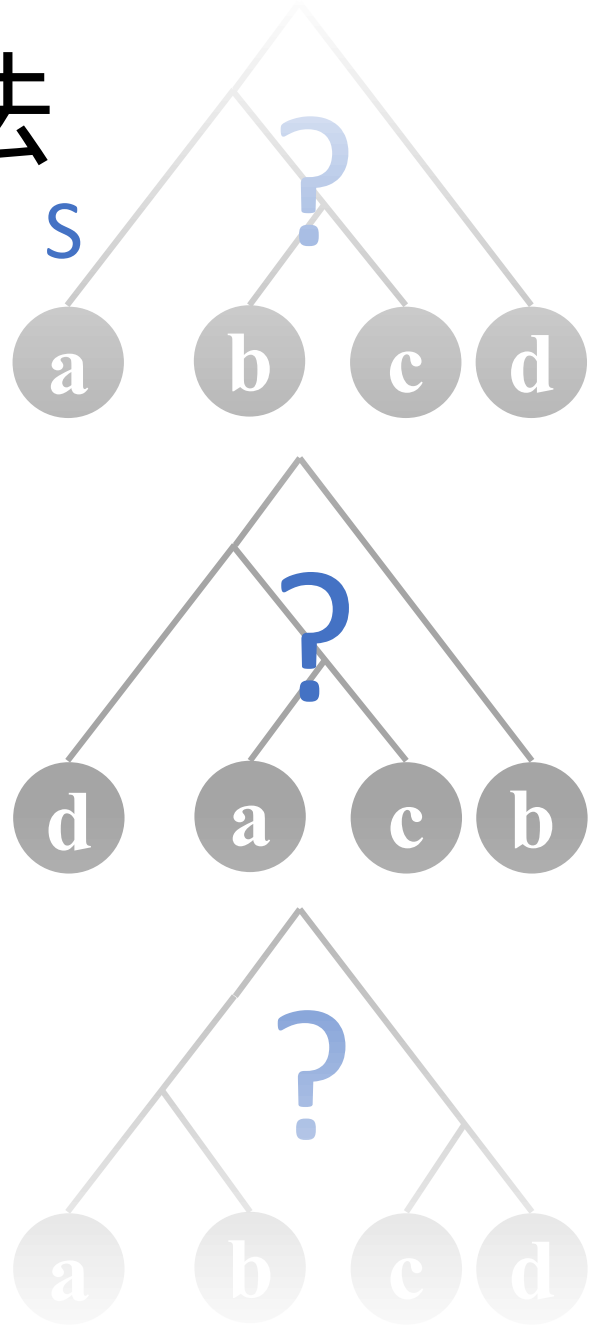


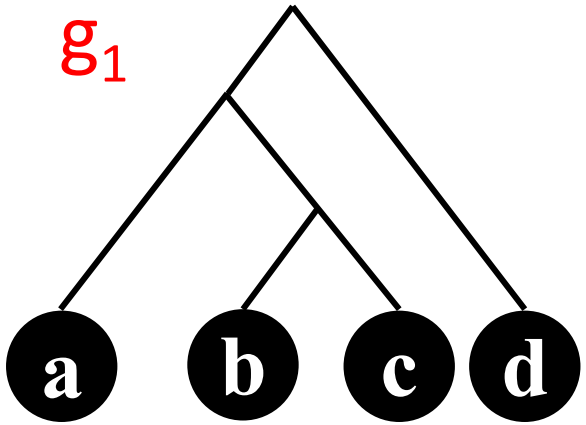
寻找**二叉树**S，使得S与g_1，…，g_n的**公共三叶树**数量之和最大化。

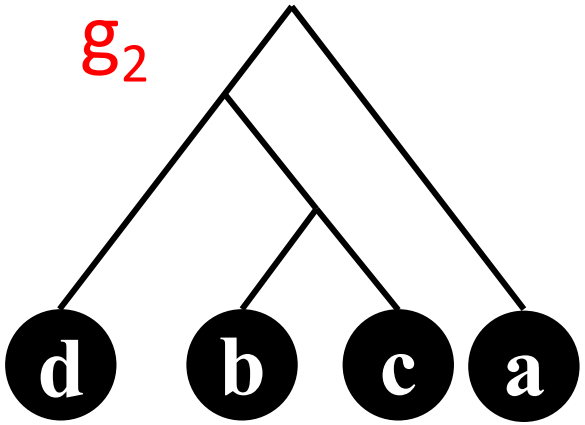


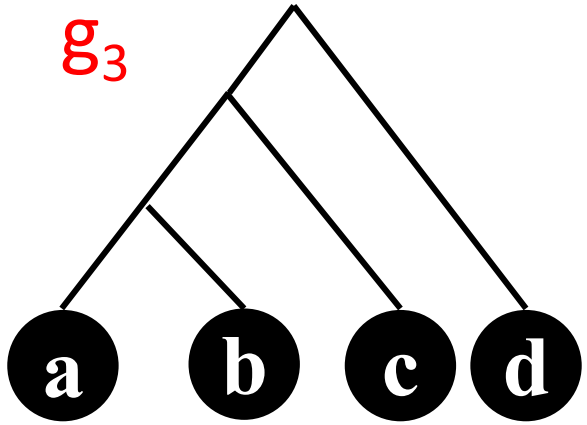


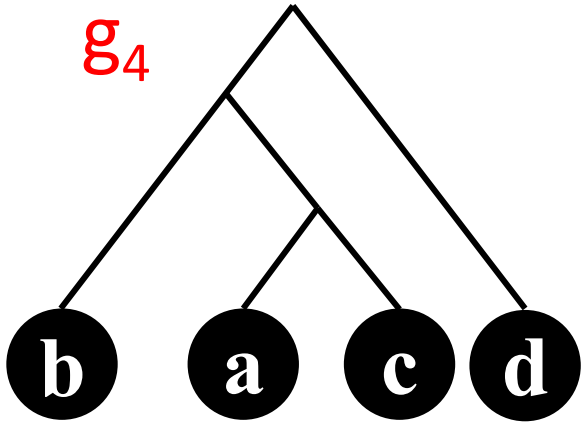
穷举法









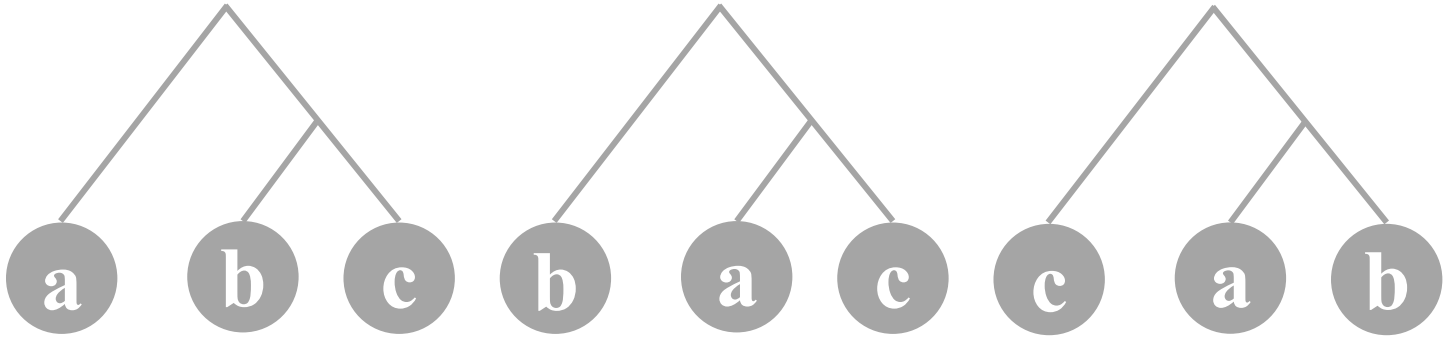




13.二叉树拓扑数量

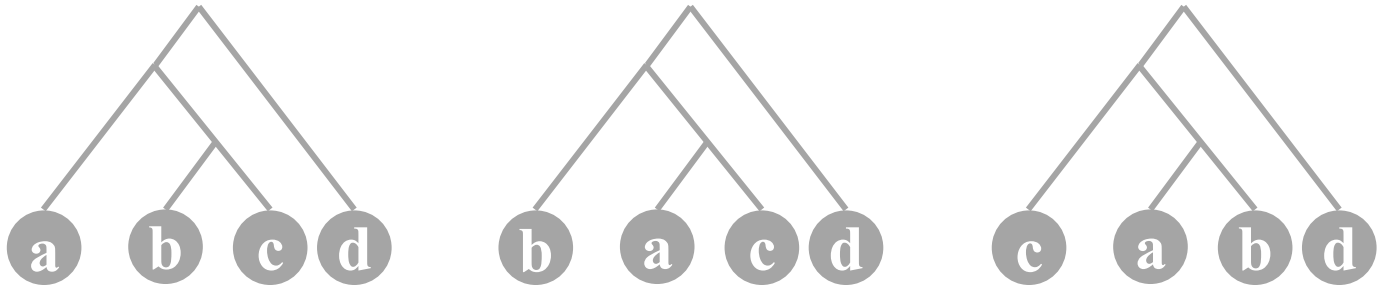
13.1 二叉树的拓扑数量

$n = 3$



3种。

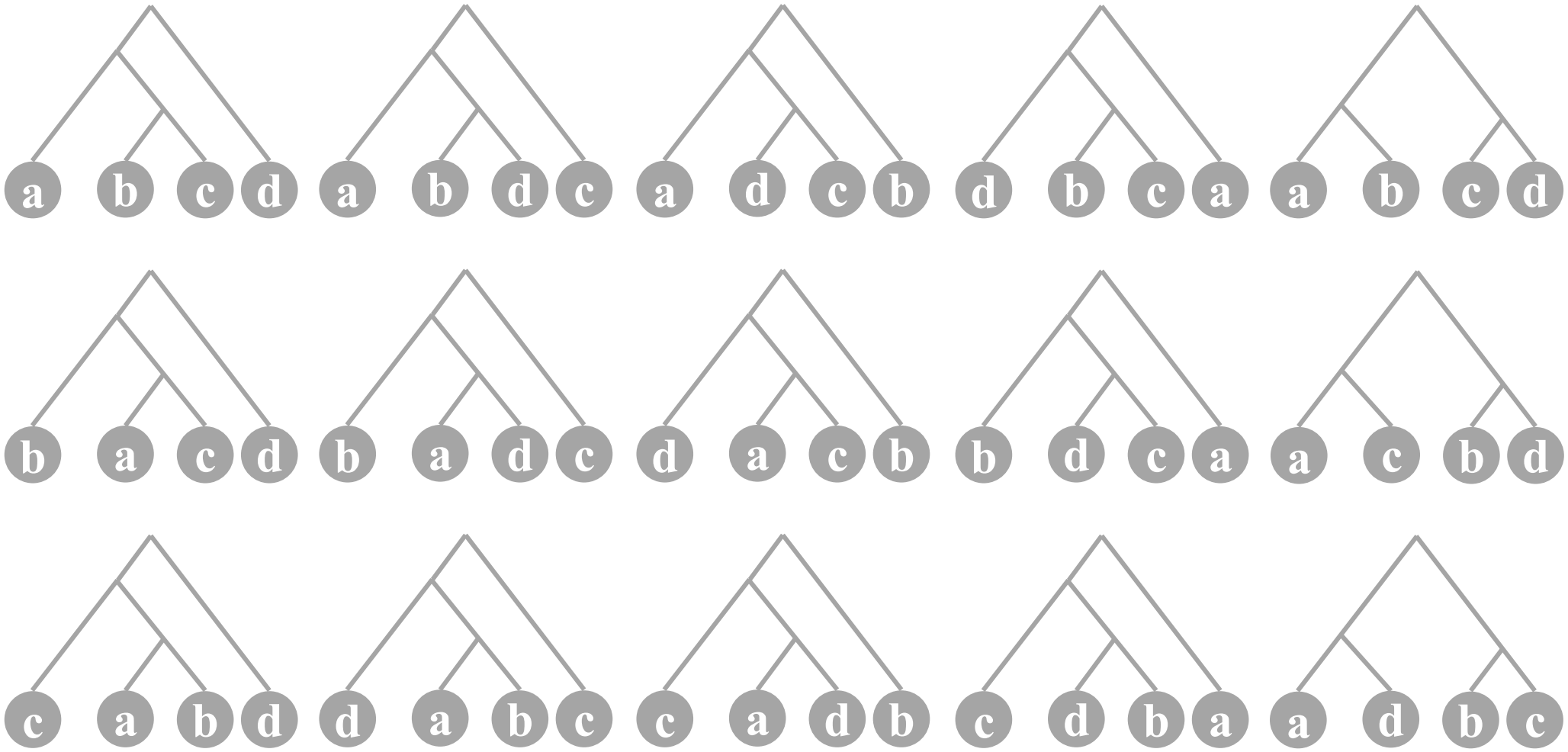
$n = 4$



几种?

二叉树的拓扑数量

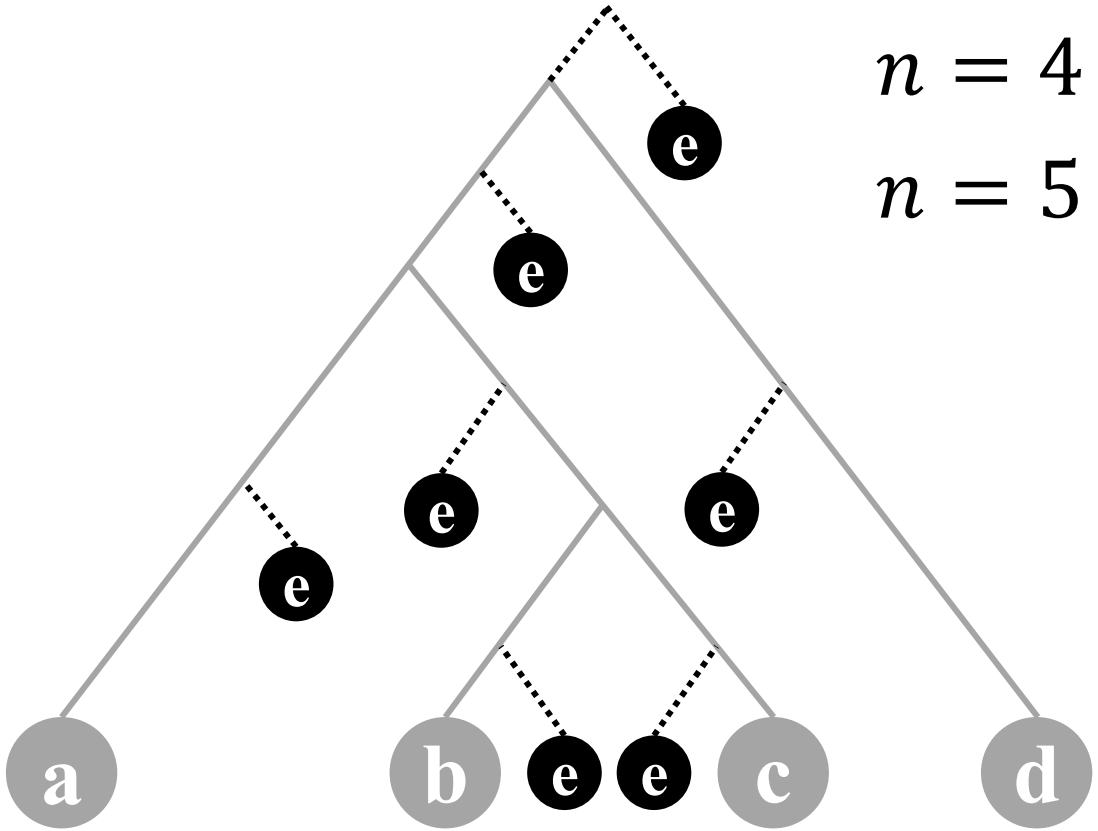
15种。



13.2 二叉树的拓扑数量

$n = 2$	1种。
$n = 3$	3种。
$n = 4$	15种。
$n = 5$	几种?
⋮	⋮
n	几种?

二叉树的拓扑数量



$n = 4$　15种。

$n = 5$　$15 \times 7 = 105$种

二叉树的拓扑数量

$n = 2$	1种。	$n = 8$	135,135种。
$n = 3$	3种。	$n = 9$	2,027,025种。
$n = 4$	15种。	$n = 10$	34,459,425种。
$n = 5$	105种。	$n = 11$	654,729,075种。
$n = 6$	945种。	$n = 12$	13,749,310,575种。
$n = 7$	10,395种。	$n = 13$	316,234,143,225种。
⋮	⋮		
n	$\prod_{i=1}^{n-1} 2i - 1$种。		



14.无根二叉树拓扑数量

14.1 无根二叉树的拓扑数量

$n = 3$ 1种。

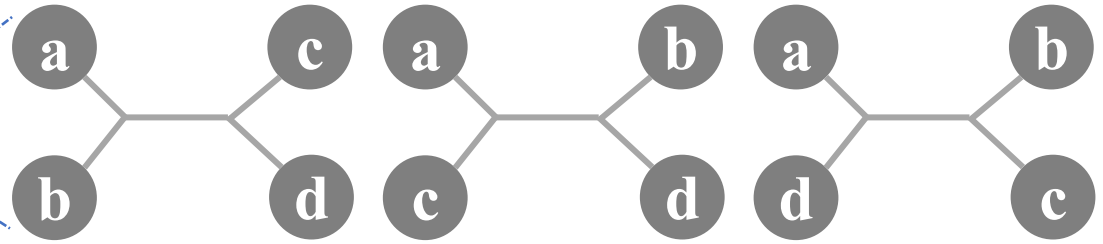
$n = 4$ 3种。

$n = 5$ 几种?

$n = 6$ 几种?

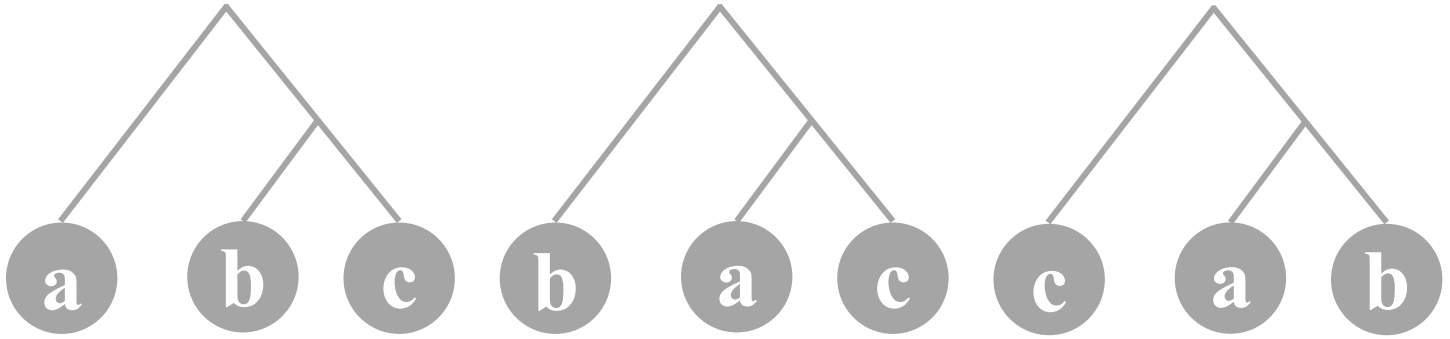
$\vdots$ $\vdots$

n 几种?



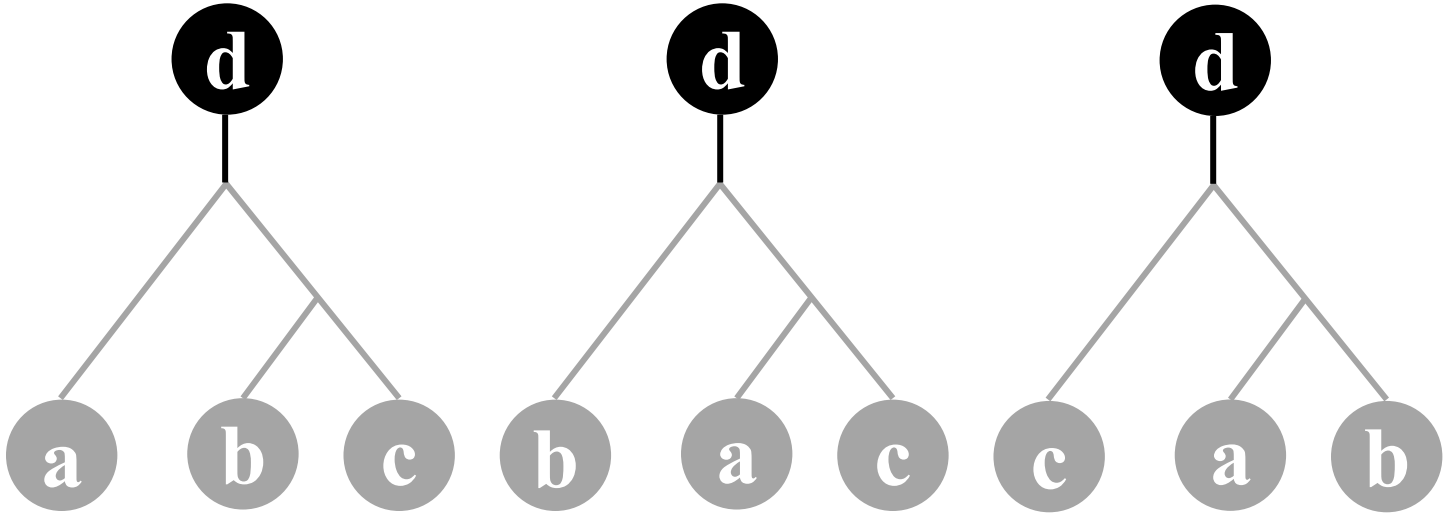
有根/无根二叉树的拓扑数量

有根
$n = 3$



3种。

无根
$n = 4$



3种。

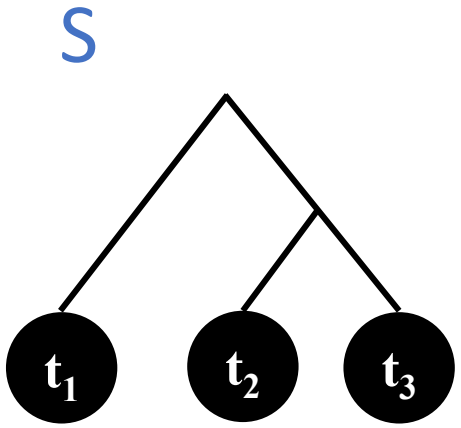
有根/无根二叉树的拓扑数量

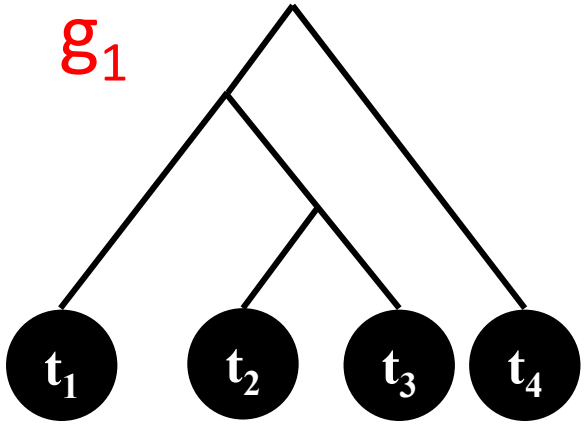
有根		无根	
$n=2$	1种。	$n=3$	1种。
$n=3$	3种。	$n=4$	3种。
$n=4$	15种。	$n=5$	15种。
$n=5$	105种。	$n=6$	105种。
$n=6$	945种。	$n=7$	945种。
⋮	⋮	⋮	⋮
n	$\prod_{i=1}^{n-1} 2i-1$种。	n	$\prod_{i=1}^{n-2} 2i-1$种。

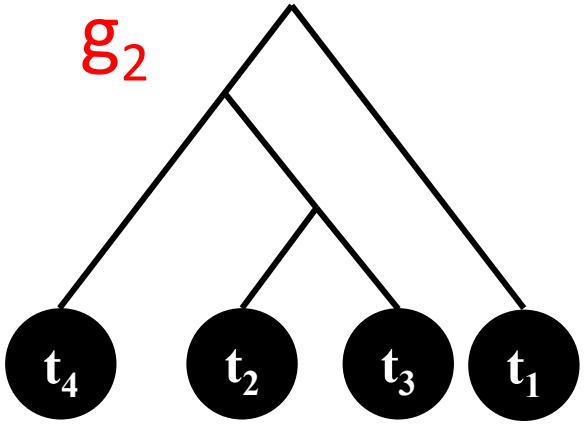


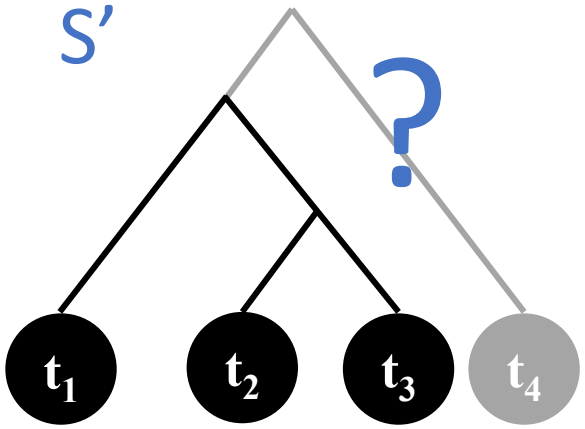
15.三叶树叶节点放置问题

三叶树叶节点放置问题





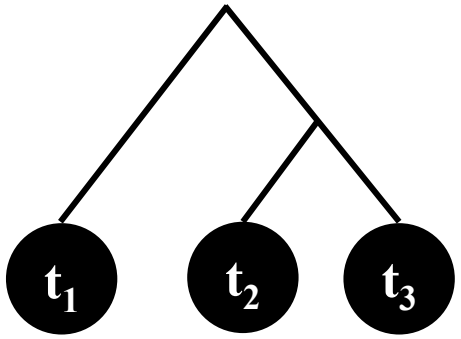




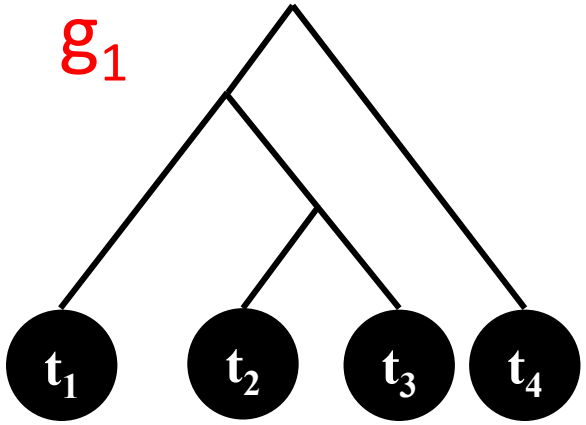
给定包含**叶子节点**t_1，…，t_{n-1}的**二叉树**S和包含**叶子节点**t_1，…，t_n的**二叉树**g_1，…，g_k。将**叶子节点**t_n置于S，使得S与g_1，…，g_k的**公共三叶树**数量之和最大化。

15.1 三叶树叶节点放置问题

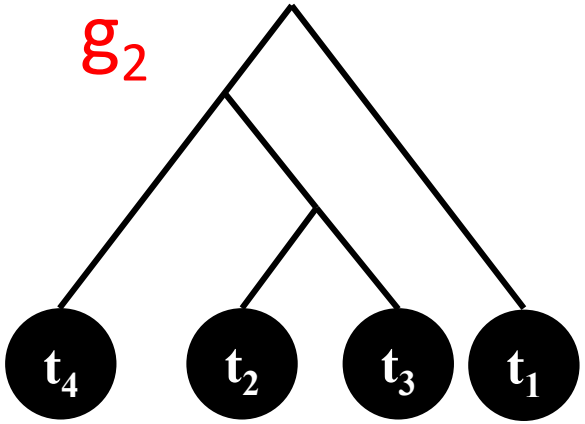
S



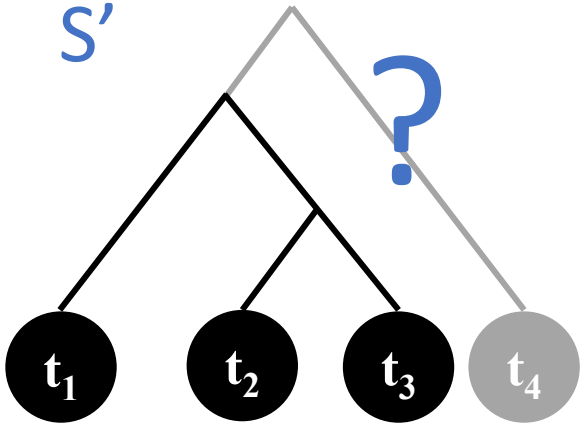
g_1



g_2



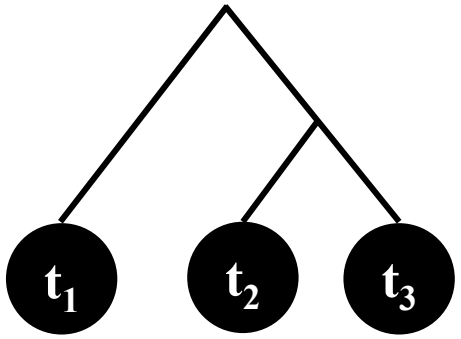
S'



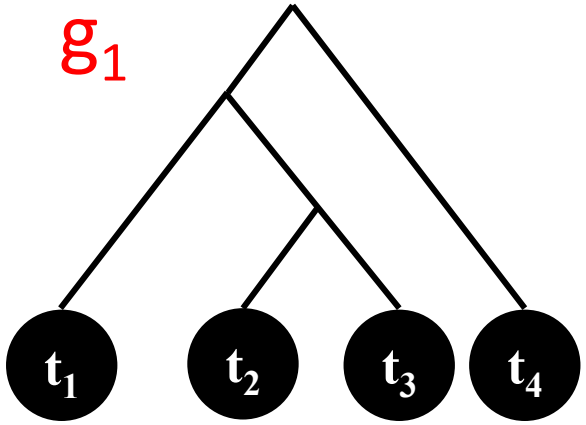
求三叶树叶节点放置问题的时间复杂度。

三叶树叶节点放置问题

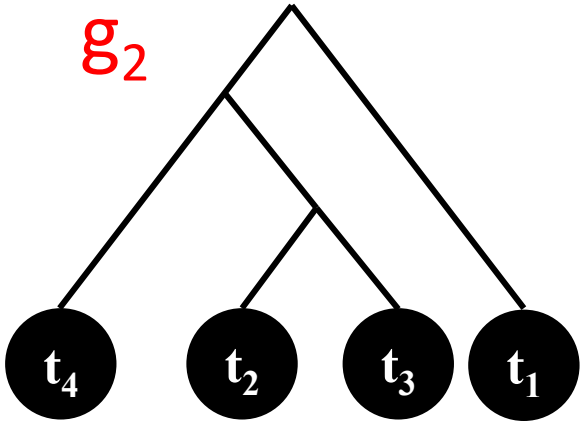
S



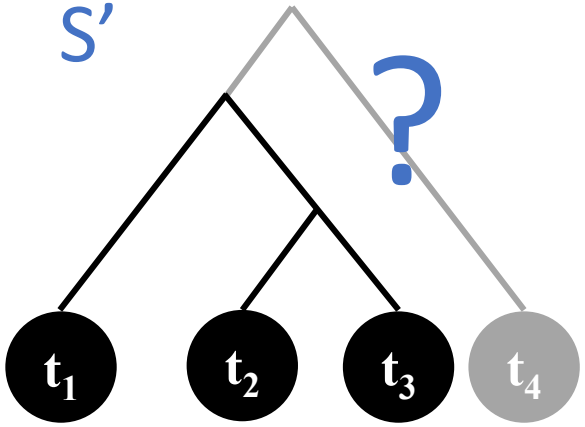
g_1



g_2



S’

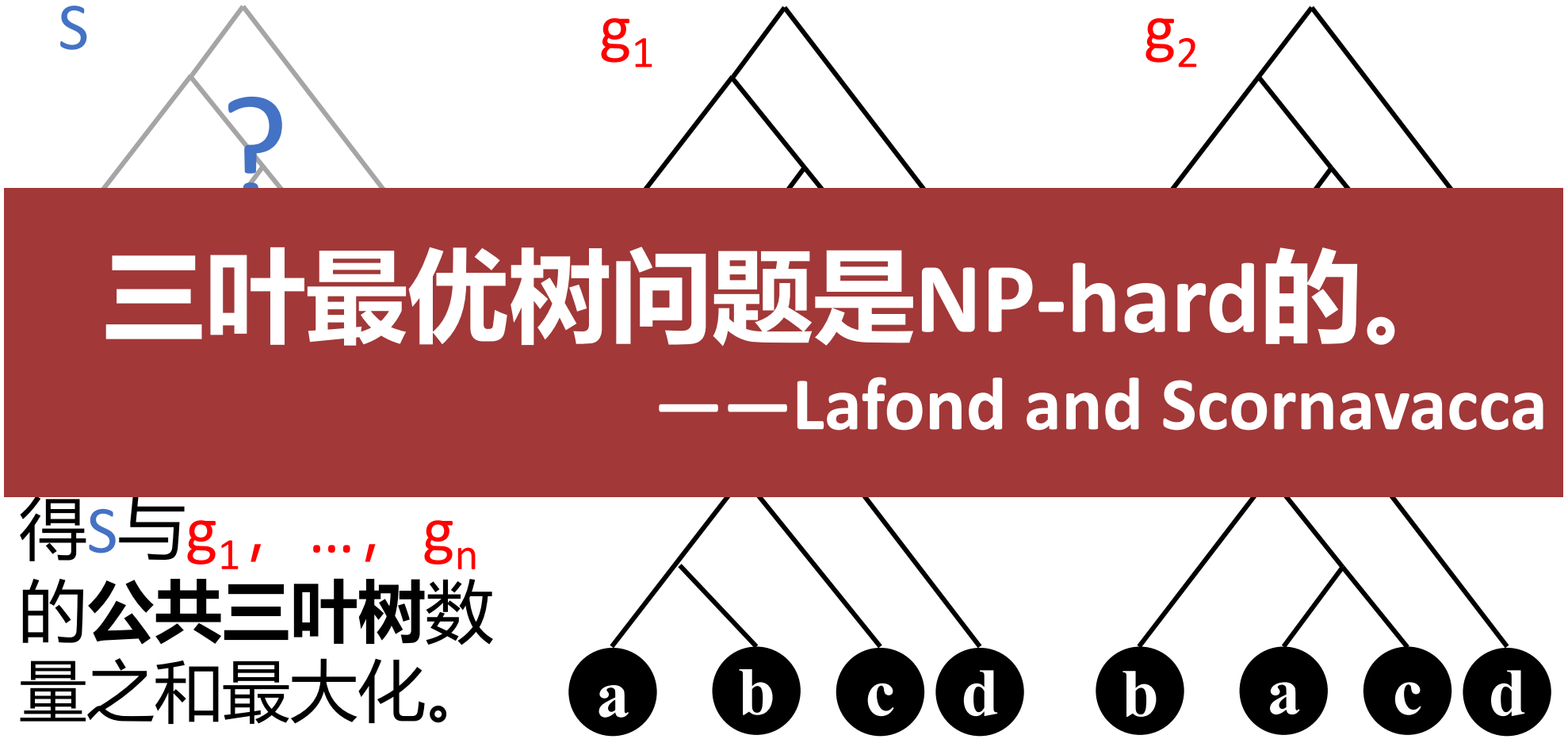


穷举法的时间复杂度为$O(n^3k)$。



16.三叶最优树问题近似解

三叶最优树问题

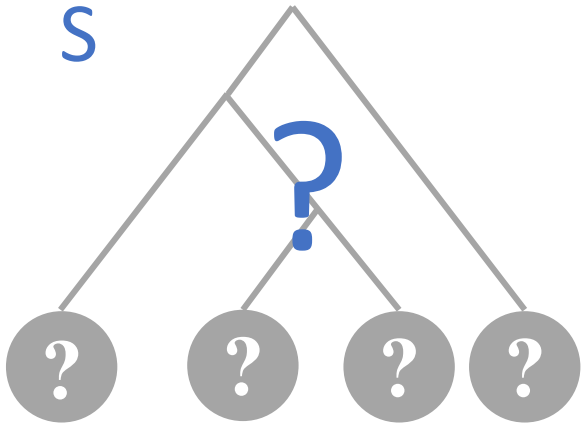


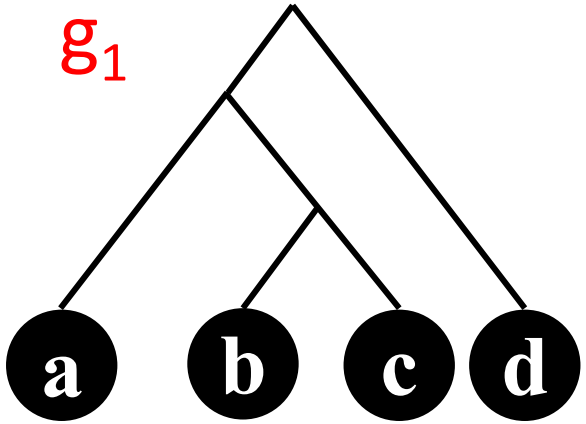
三叶最优树问题是NP-hard的。

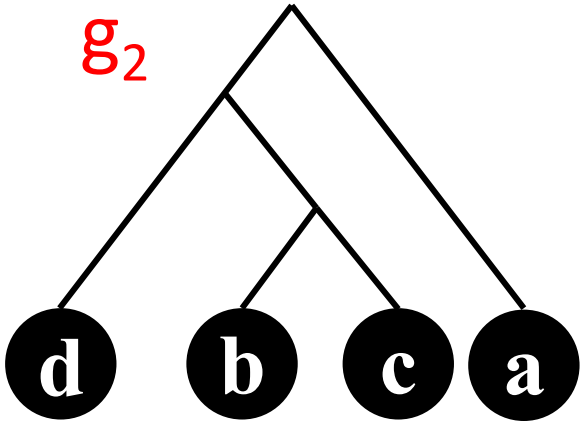
——Lafond and Scornavacca

得S与g_1，…，g_n的**公共三叶树**数量之和最大化。

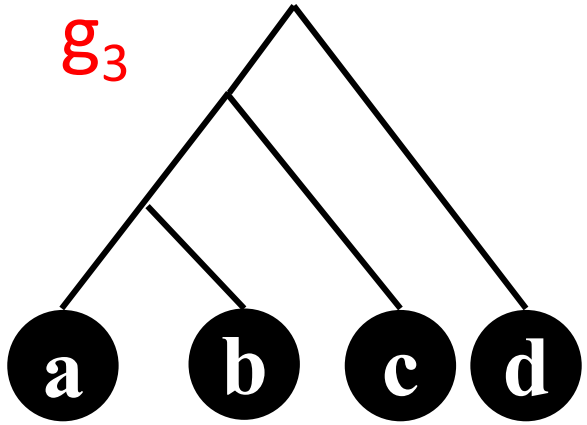
16.1 三叶最优树问题近似算法

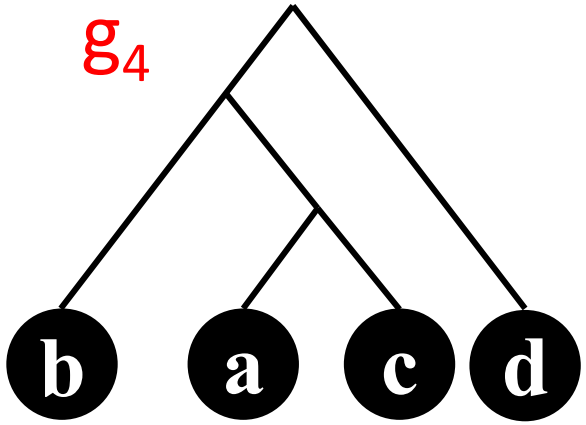




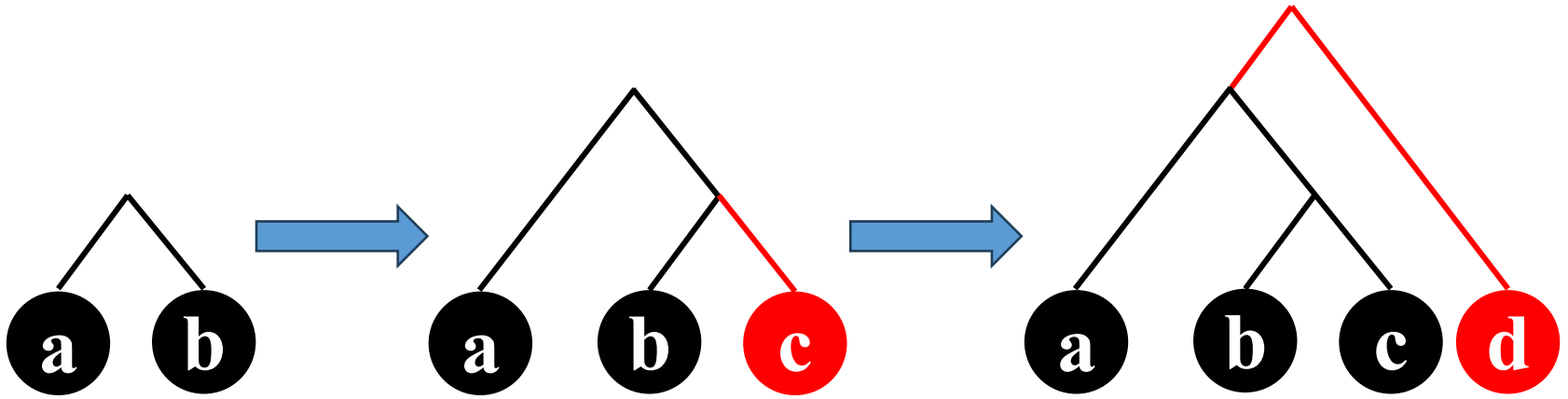


寻找**二叉树**S，使得S与g_1，…，g_n的**公共三叶树**数量之和尽量大化。





16.2 贪心算法

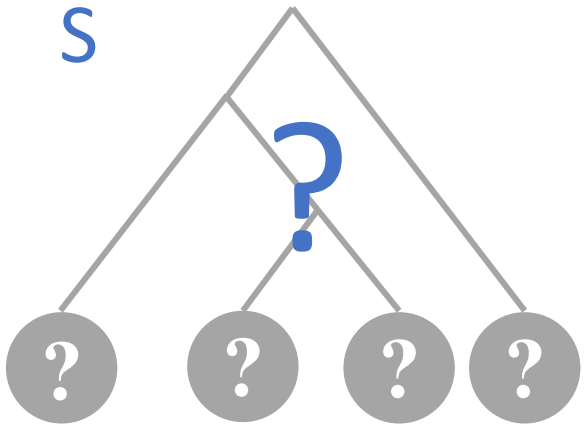


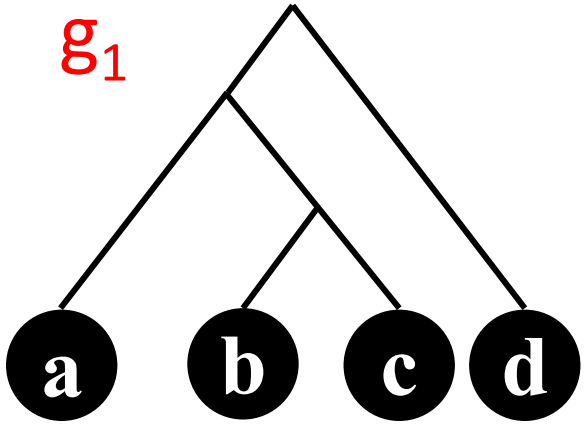
贪心算法的时间复杂度为$O(n^4k)$。
为什么该算法无法保证得到最优解?

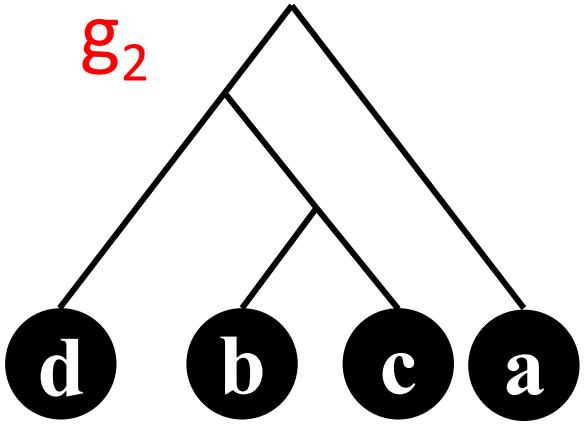


25.双色三叶非二叉树问题

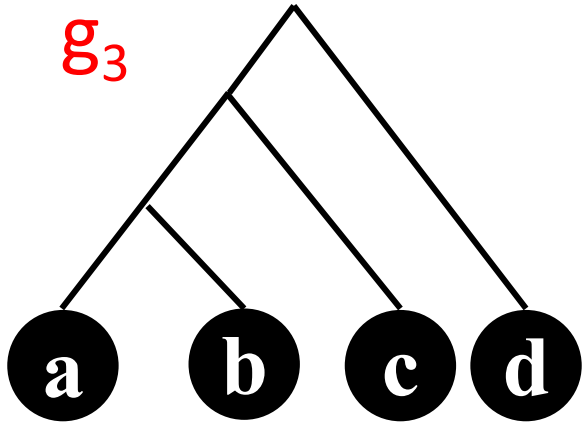
三叶最优树问题

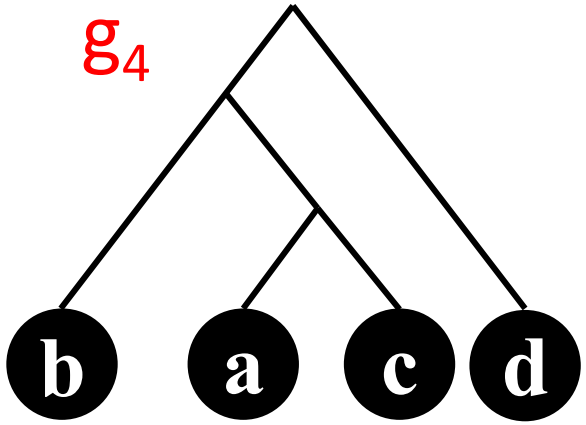




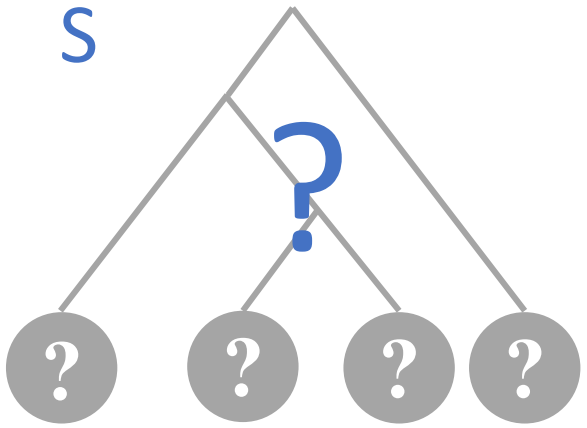


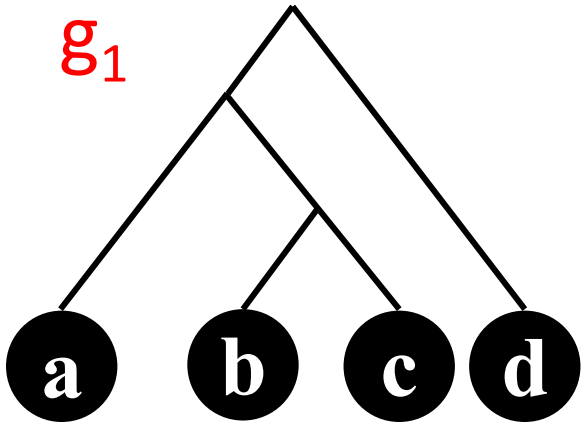
寻找**二叉树**S，使得S与g_1，…，g_n的**公共三叶树**数量之和最大化。

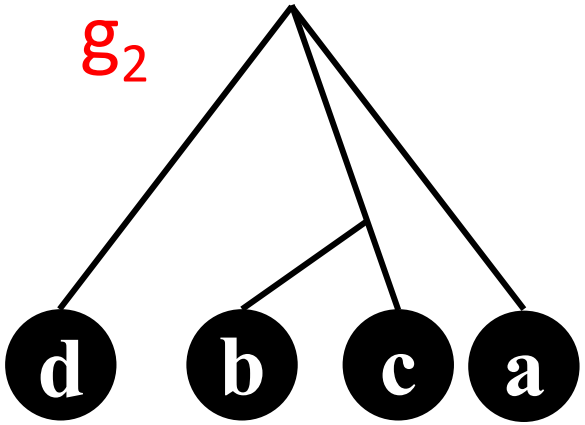




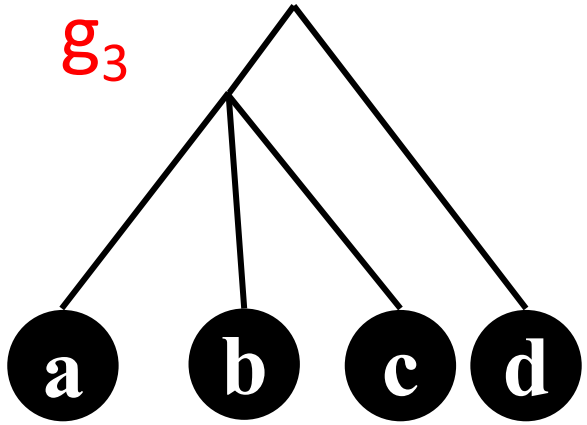
三叶最优树问题

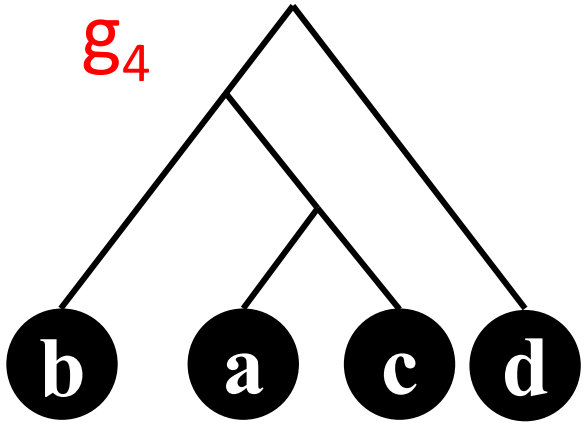




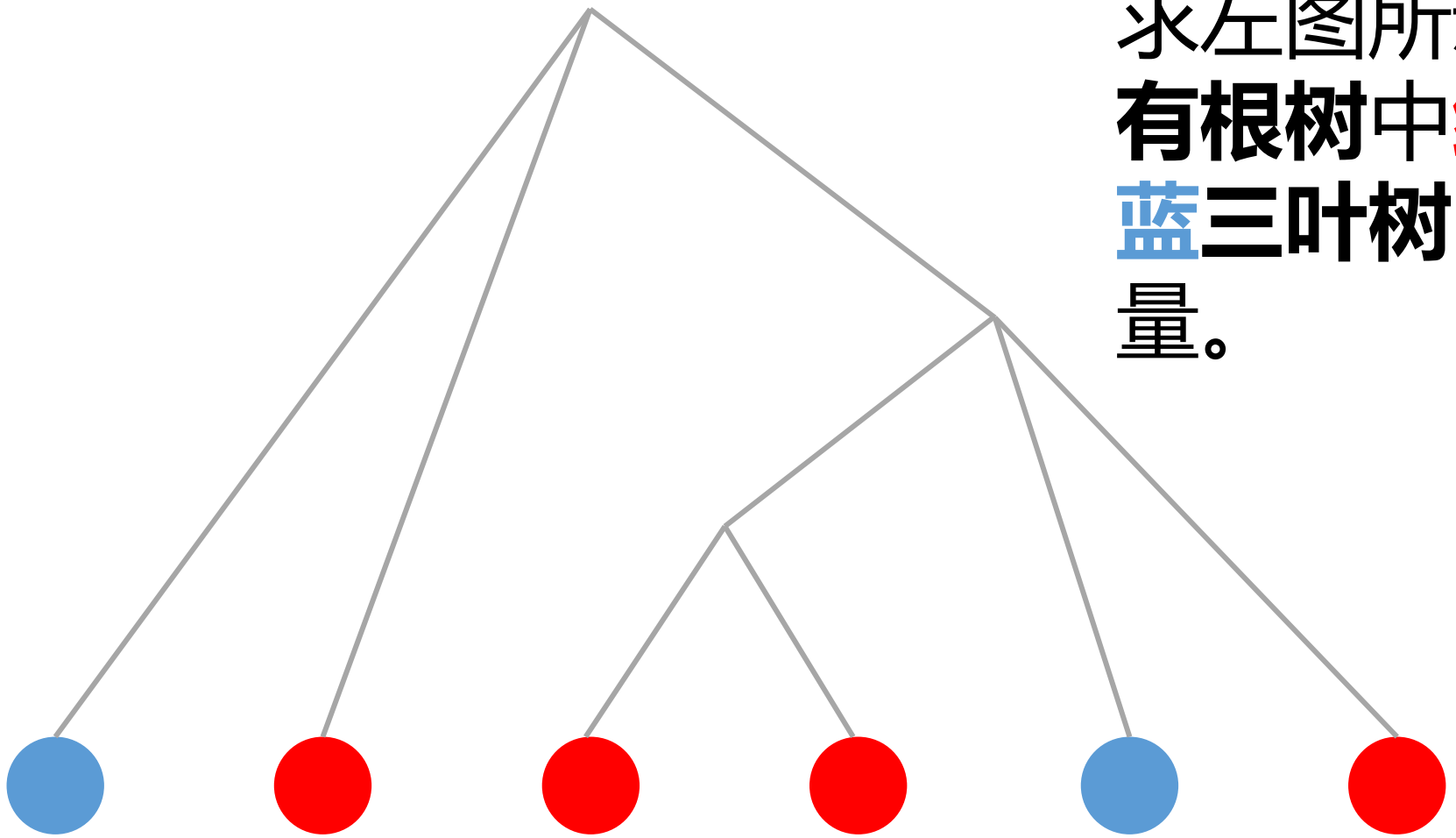


寻找**二叉树**S，使得S与g_1，…，g_n的**公共三叶树**数量之和最大化。



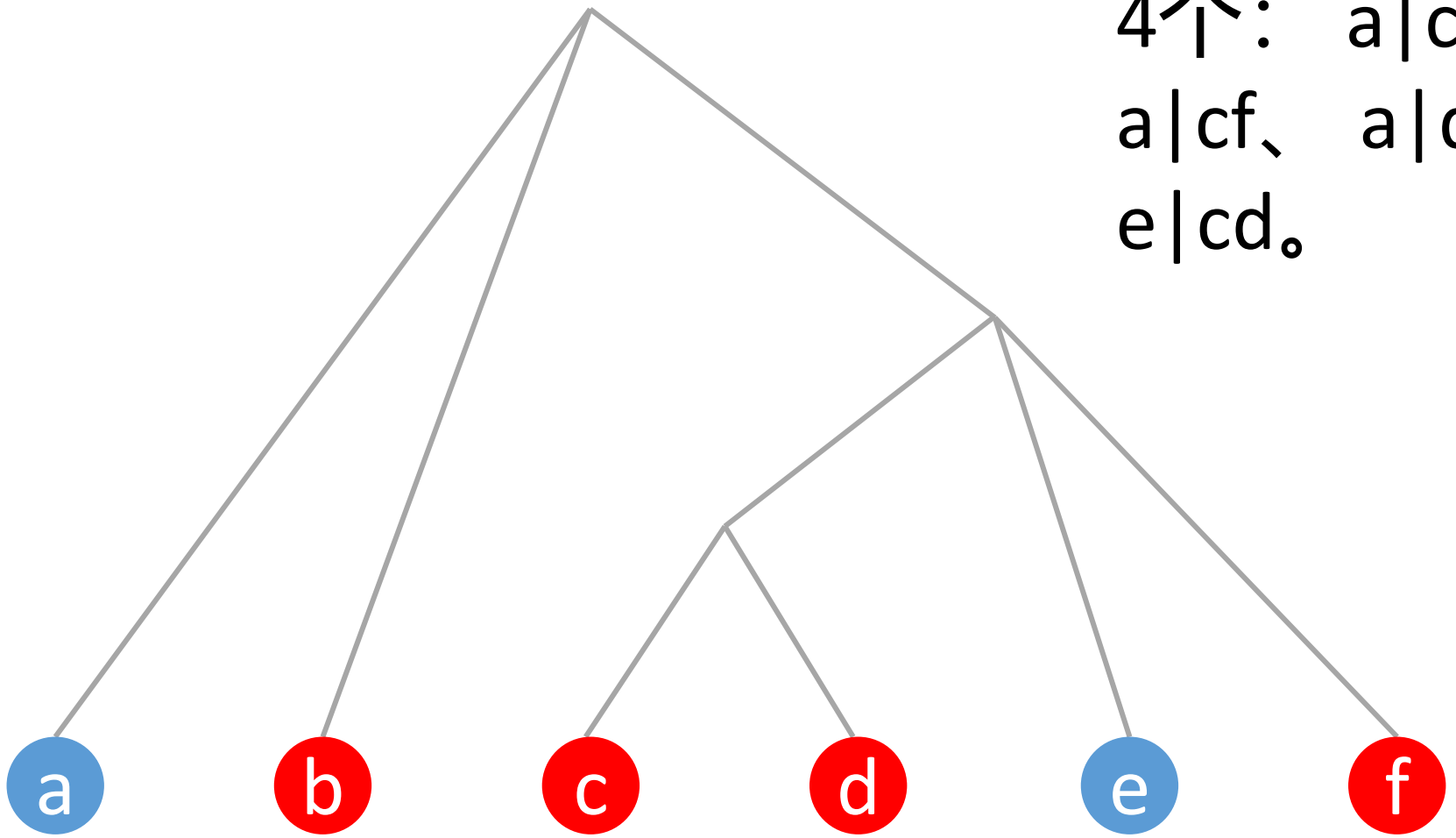


25.1 红红蓝三叶树数量



求左图所示的**有根树**中**红红蓝三叶树**的数量。

红红蓝三叶树数量



4个： a|cd、a|cf、 a|df、e|cd。

25.2 红红蓝三叶树算法

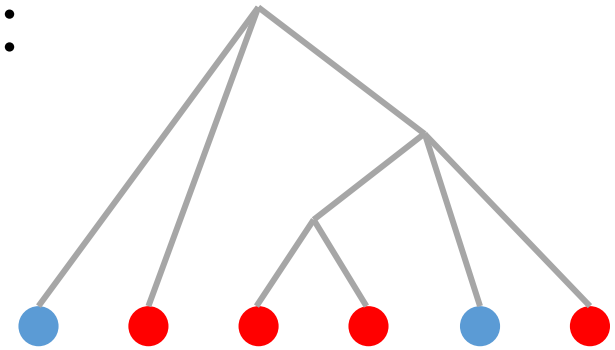
输入：给定的包含n个叶子节点的有根树，其叶子节点被染成红色或蓝色。

输出：该有根树中**红红蓝三叶树**的个数

要求：时间复杂度O(n)

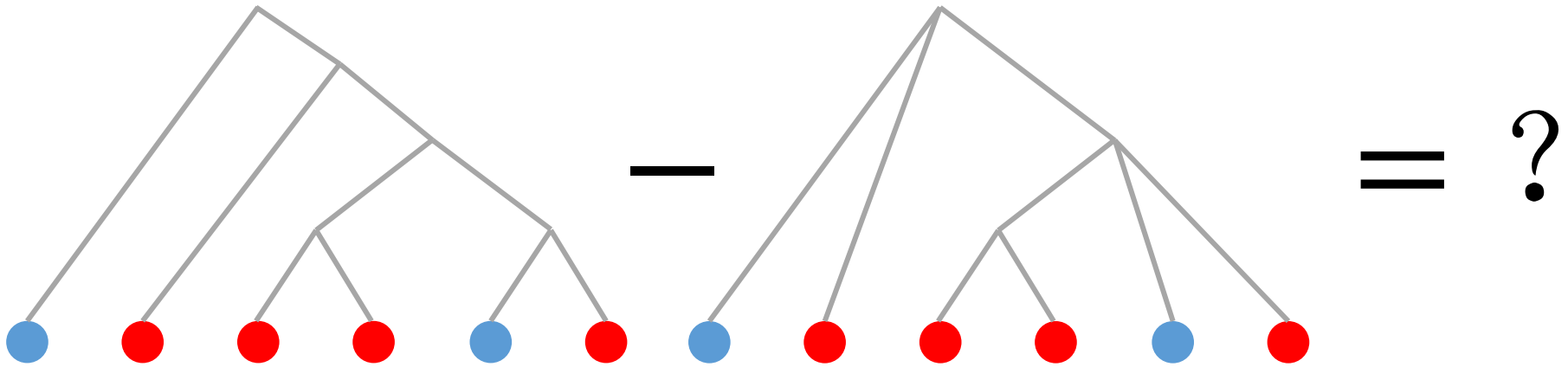
样例：

输入：

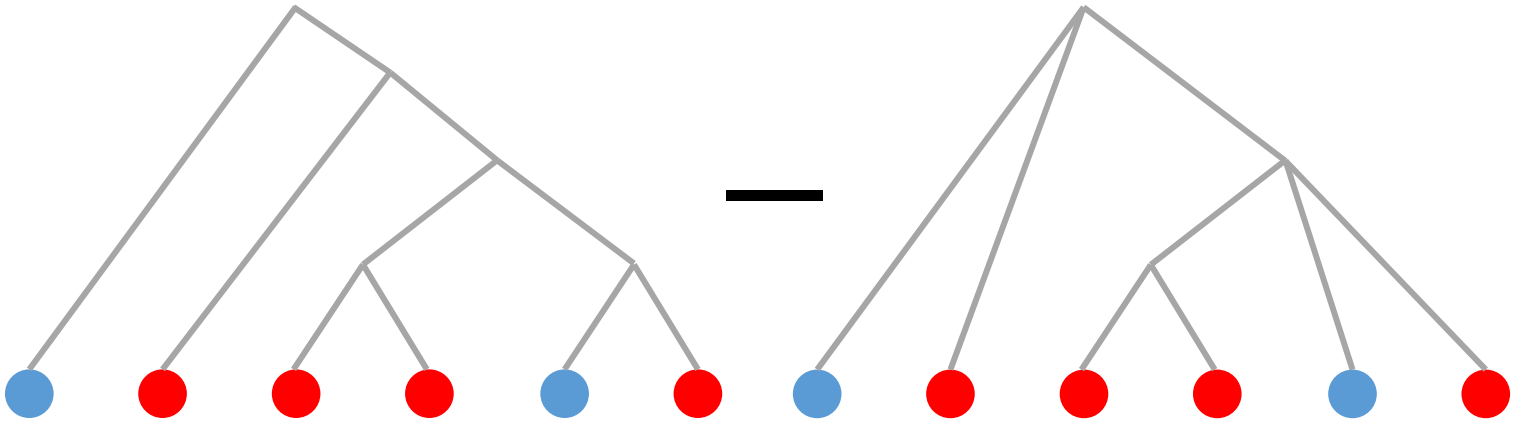


输出：4

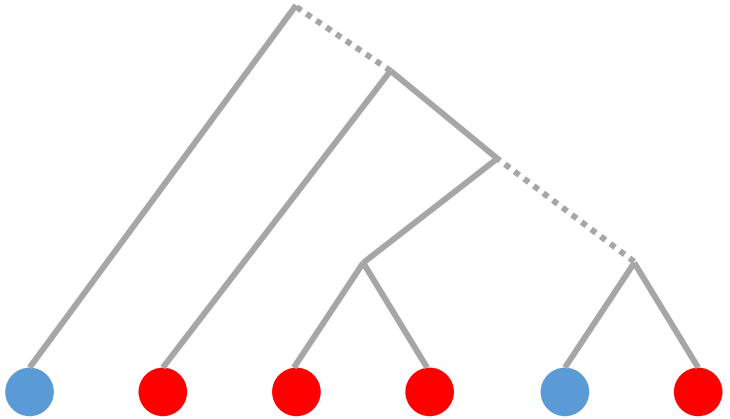
25.3 红红蓝三叶树算法



红红蓝三叶树算法



=



两个**锚点**之间
只有虚线边的
红红蓝三叶树

25.4 红红蓝三叶树算法

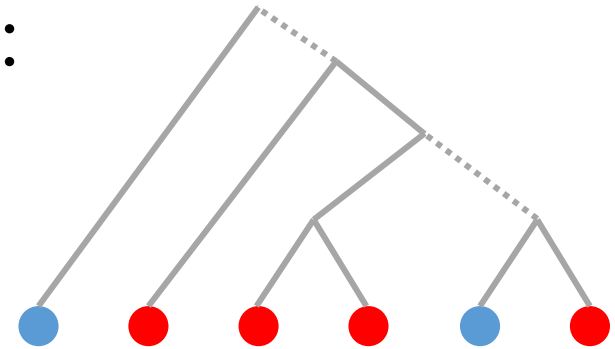
输入：给定的包含n个叶子节点的有根二叉树，其叶子节点被染成红色或蓝色。

输出：该有根二叉树中两个**锚点**之间只有虚线边的**红红蓝三叶树**的个数

要求：时间复杂度O(n)

样例：

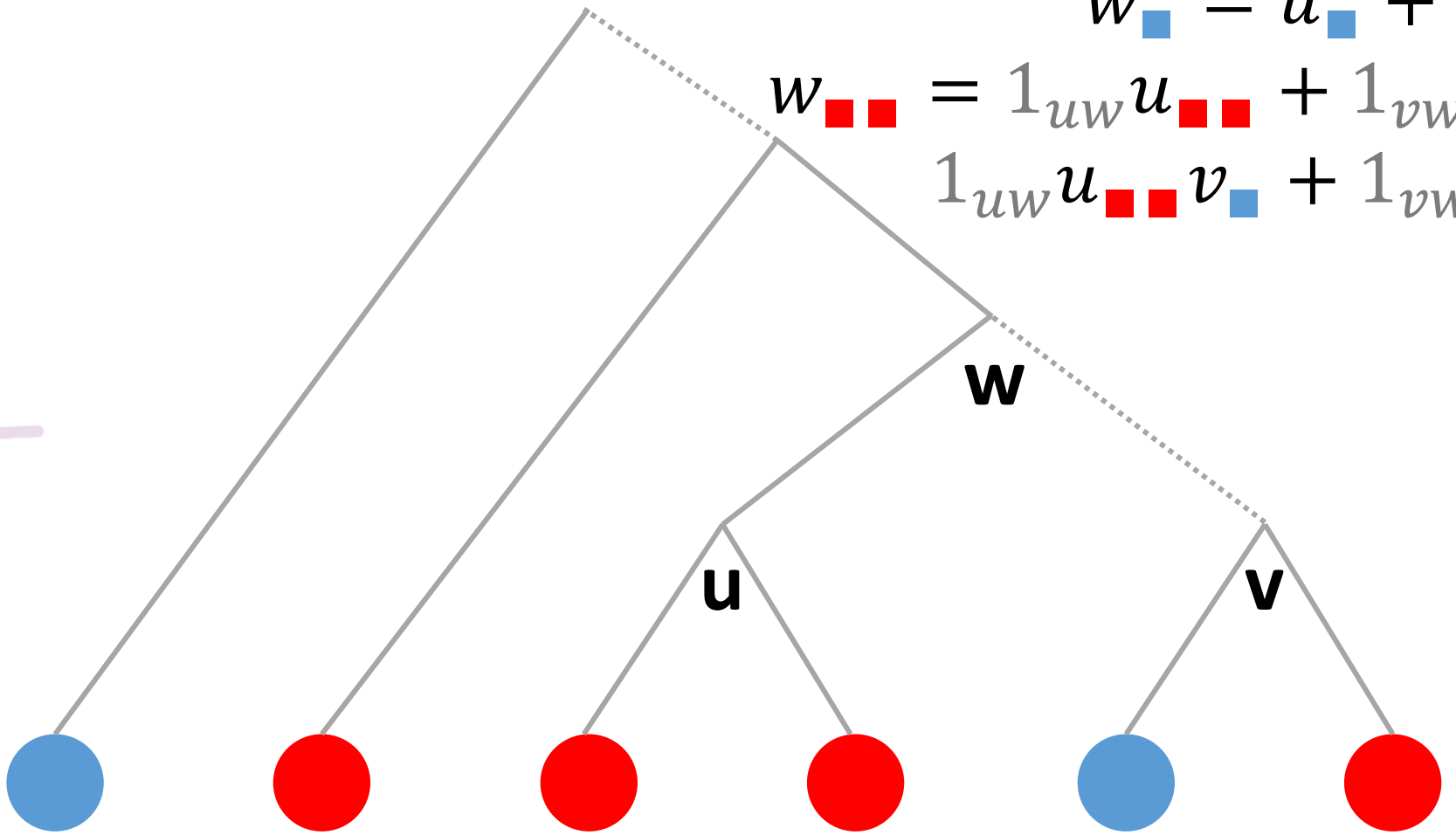
输入：



输出：3

红红蓝三叶树算法

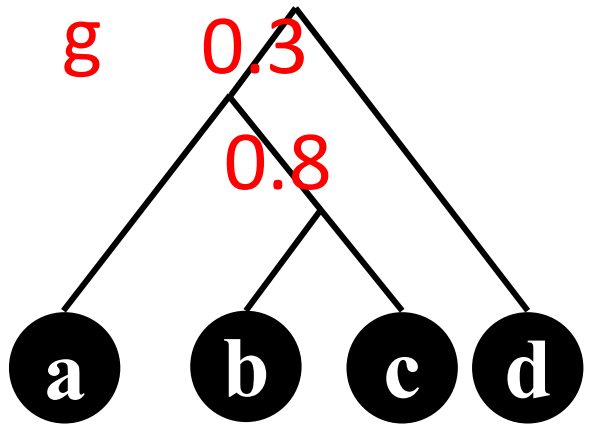
$$w_{\blacksquare} = u_{\blacksquare} + v_{\blacksquare}$$
$$w_{\blacksquare} = u_{\blacksquare} + v_{\blacksquare}$$
$$w_{\blacksquare\blacksquare} = 1_{uw} u_{\blacksquare\blacksquare} + 1_{vw} v_{\blacksquare\blacksquare} + u_{\blacksquare} v_{\blacksquare}$$
$$1_{uw} u_{\blacksquare\blacksquare} v_{\blacksquare} + 1_{vw} v_{\blacksquare\blacksquare} u_{\blacksquare}$$

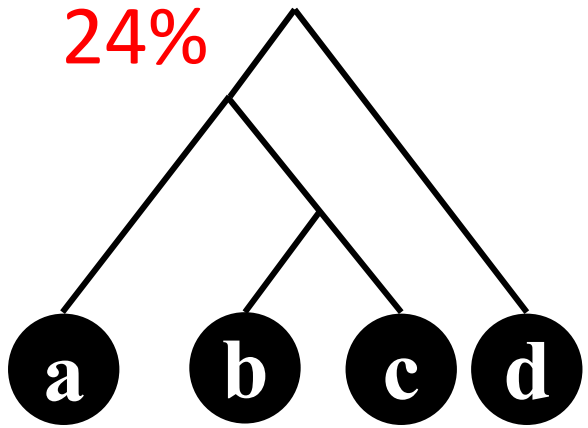


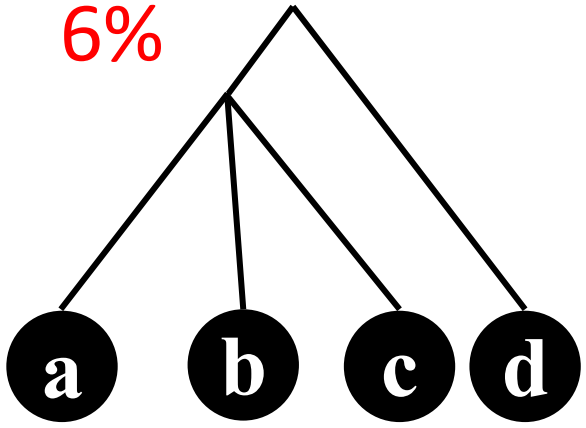


26.双色三叶支持度加权树问题

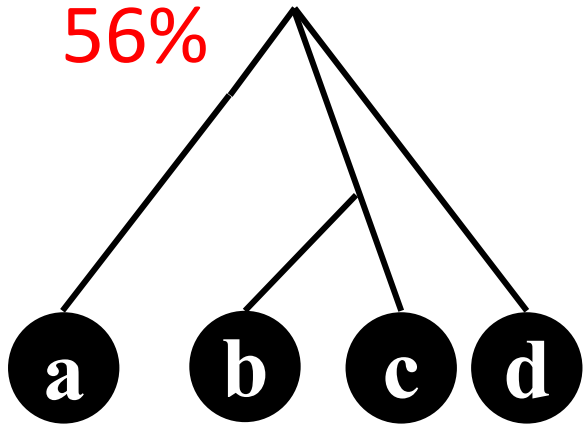
树的支持度与分布

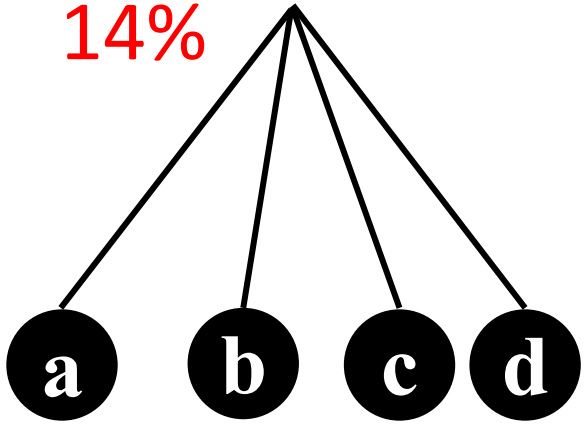




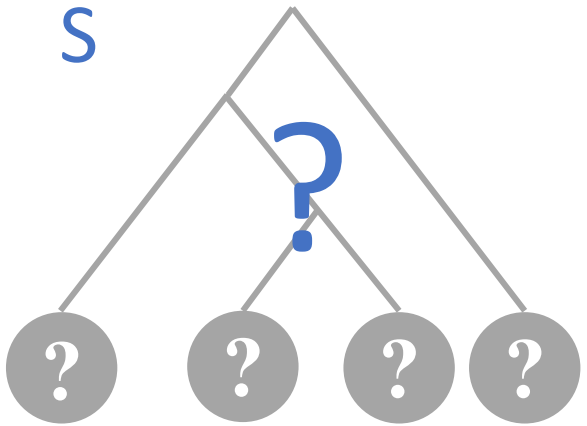


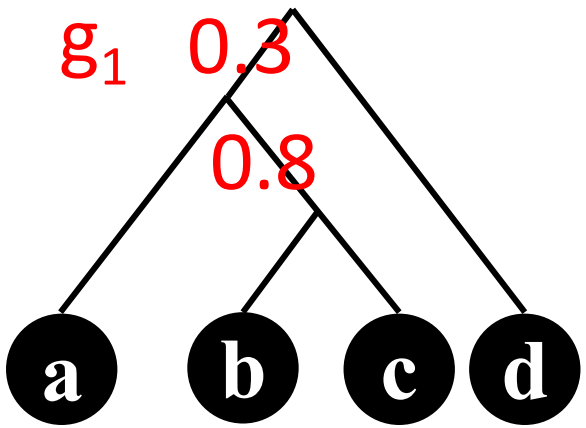
g是一个**概率分布**，其每个**内部枝**的**支持度**代表该**枝**存在的**概率**。每个**枝**存在的**事件相互独立**。

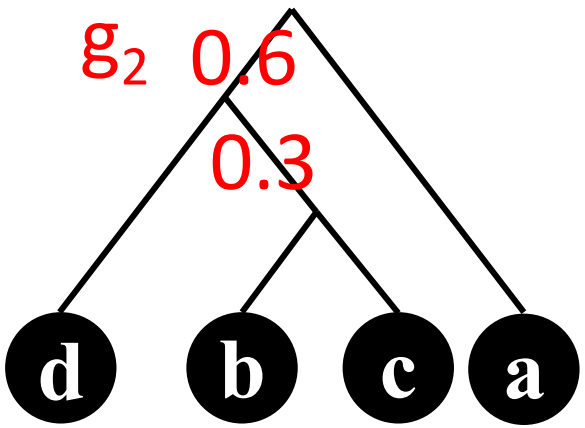




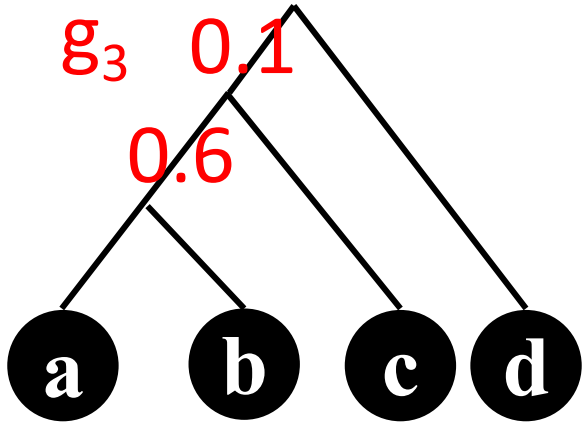
三叶最优树问题

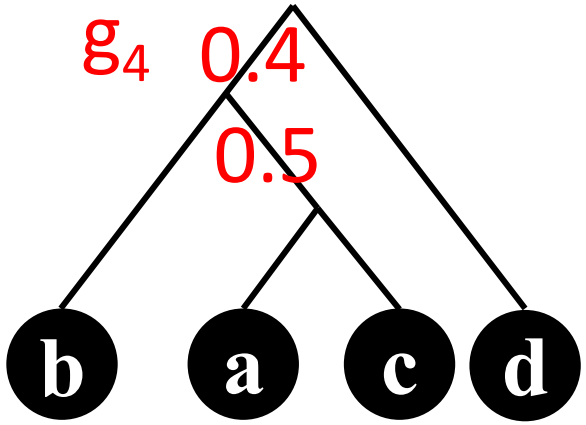




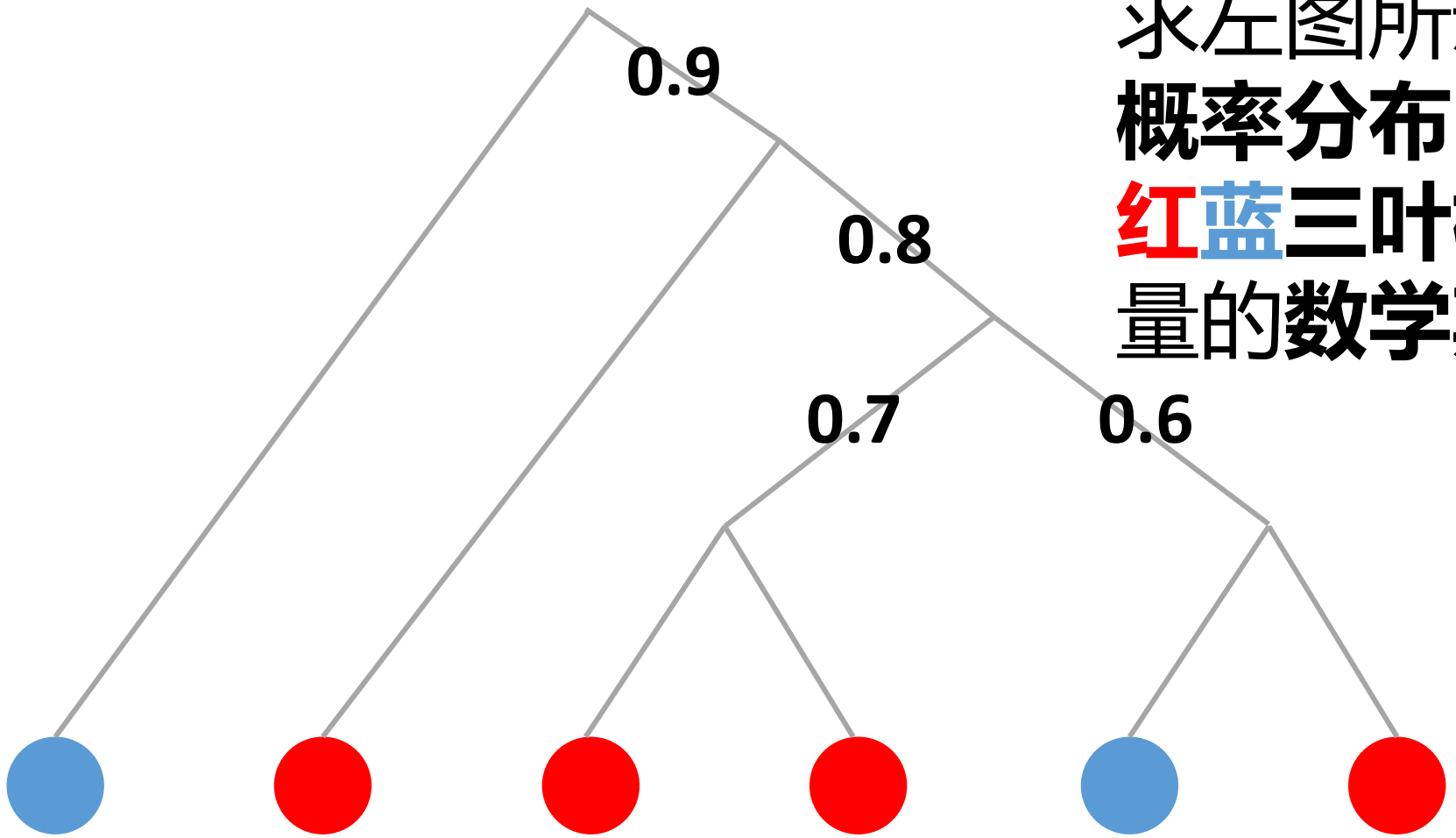


寻找**二叉树**S，使得S与g_1，…，g_n的**公共三叶树**数量之和的**数学期望**最大化。





26.1 红红蓝三叶树数量期望



求左图所示的**概率分布**中**红红蓝三叶树**数量的**数学期望**。

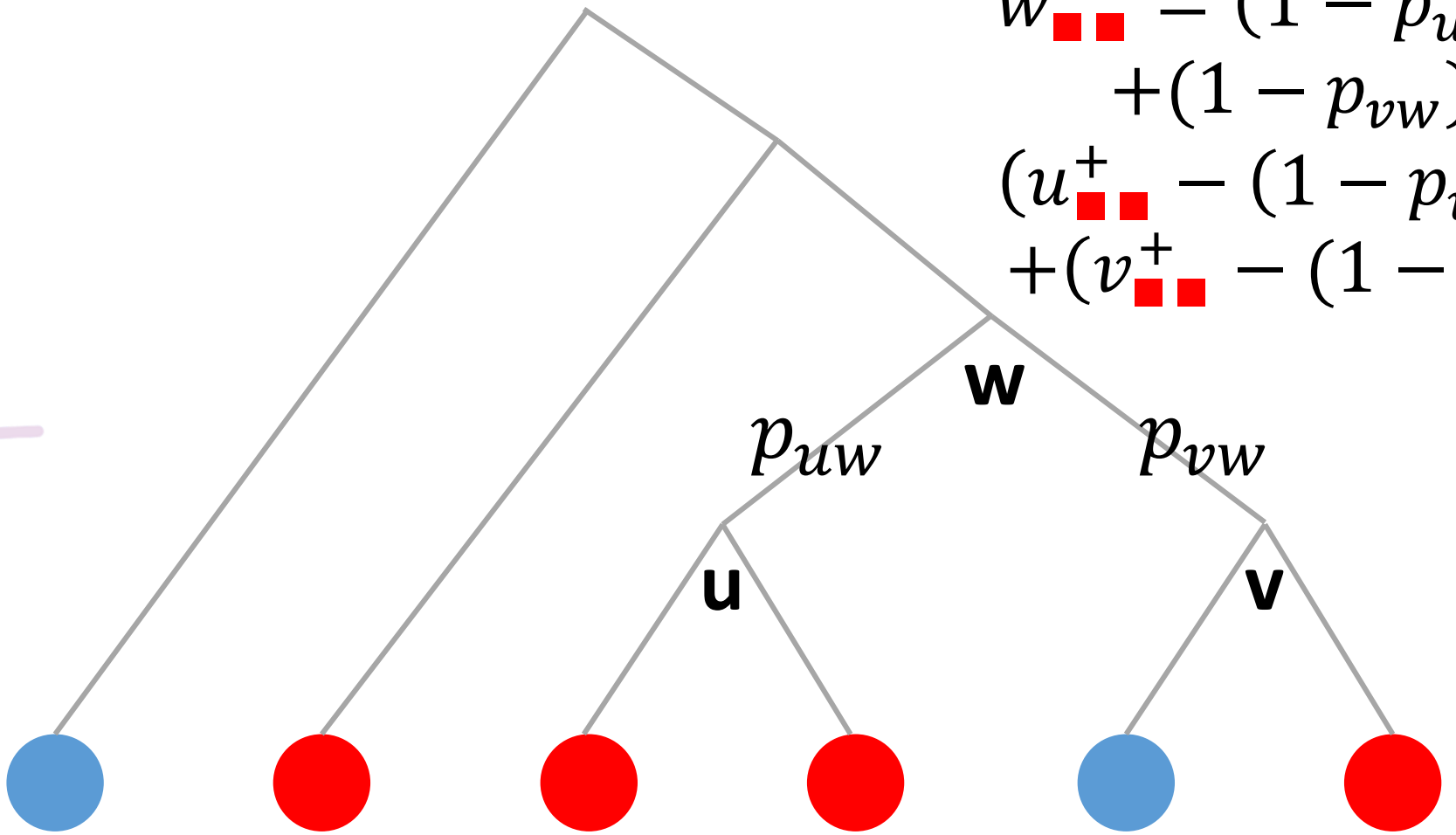
红红蓝三叶树算法

$$w^+_{\blacksquare\blacksquare} = u^+_{\blacksquare\blacksquare} + v^+_{\blacksquare\blacksquare} + u_{\blacksquare} v_{\blacksquare}$$

$$w^-_{\blacksquare\blacksquare} = (1 - p_{uw}) u^-_{\blacksquare\blacksquare} + (1 - p_{vw}) v^-_{\blacksquare\blacksquare} + u_{\blacksquare} v_{\blacksquare}$$

$$(u^+_{\blacksquare\blacksquare} - (1 - p_{uw}) u^-_{\blacksquare\blacksquare}) v_{\blacksquare}$$

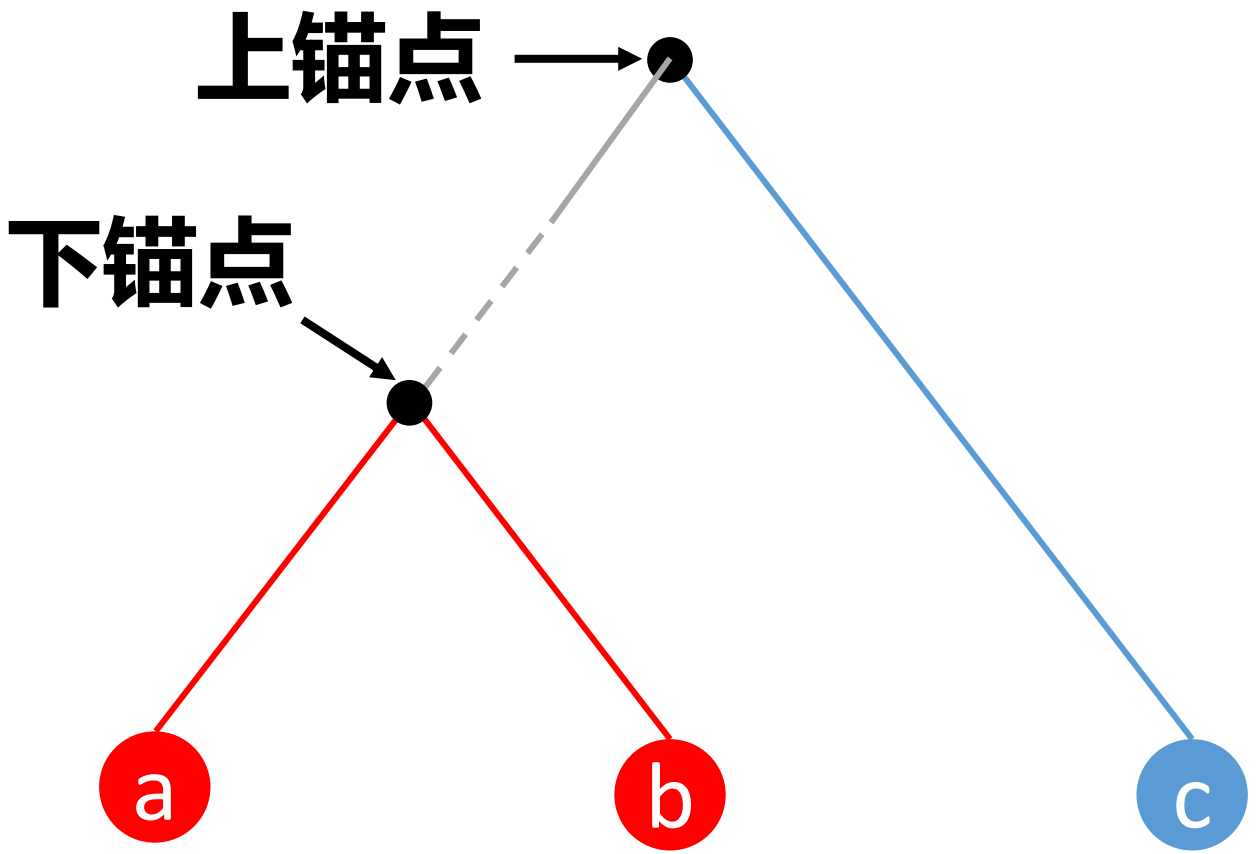
$$+ (v^+_{\blacksquare\blacksquare} - (1 - p_{vw}) v^-_{\blacksquare\blacksquare}) u_{\blacksquare}$$





27.双色可靠三叶树问题

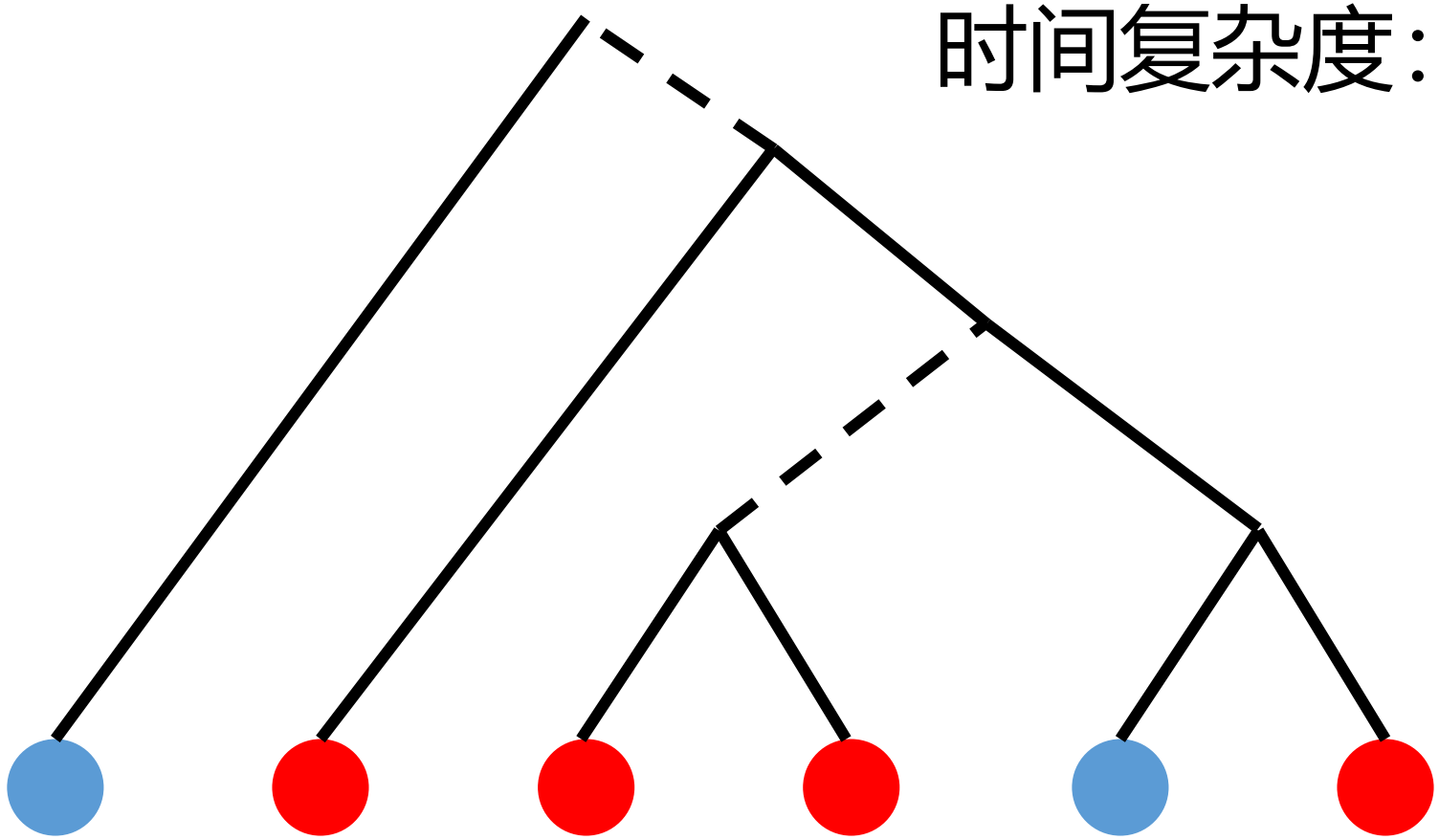
可靠三叶树



末端枝仅由实线组成的三叶树为可靠三叶树。

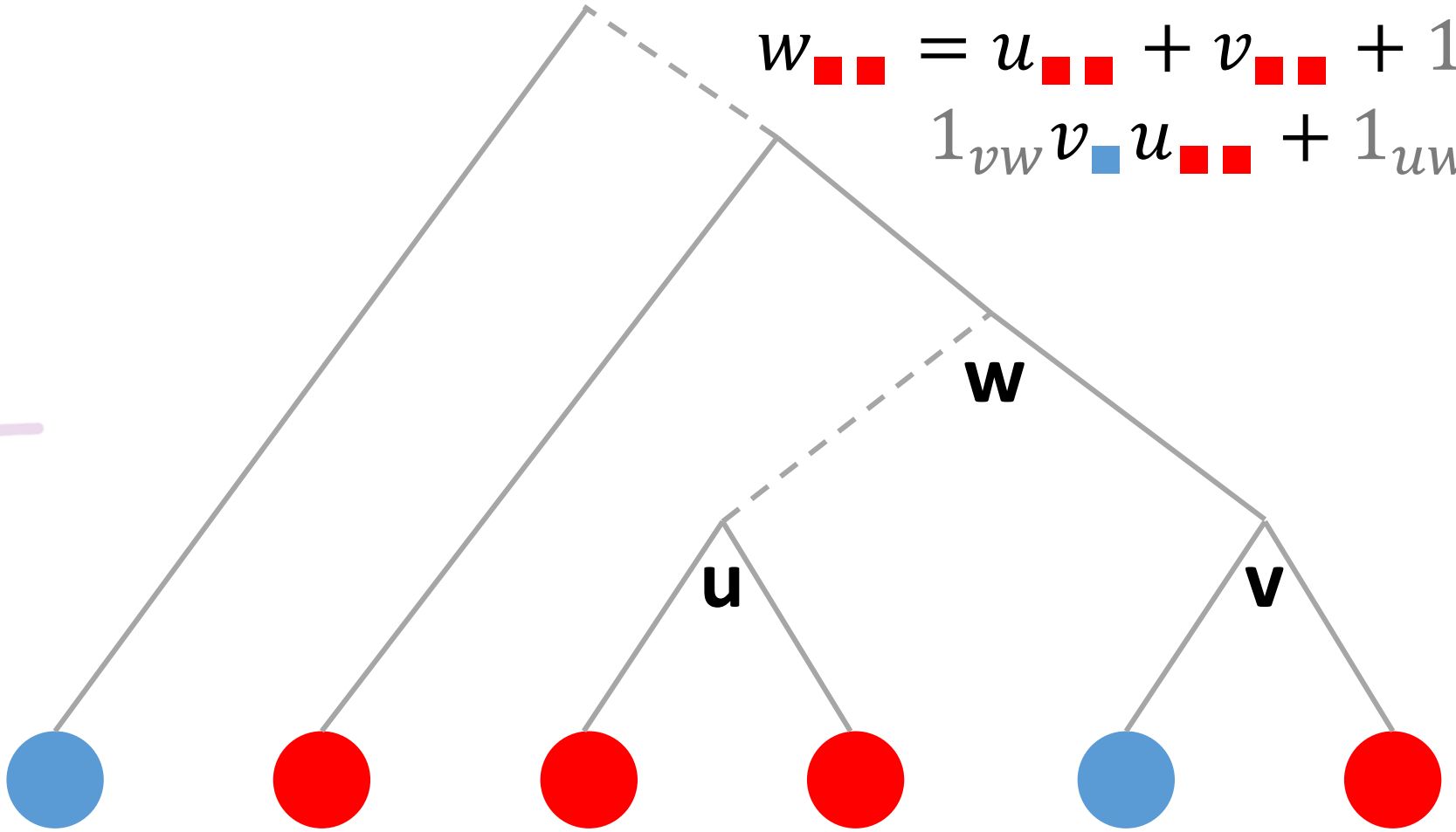
27.1 红红蓝可靠三叶树数量

时间复杂度：$O(n)$



红红蓝三叶树算法

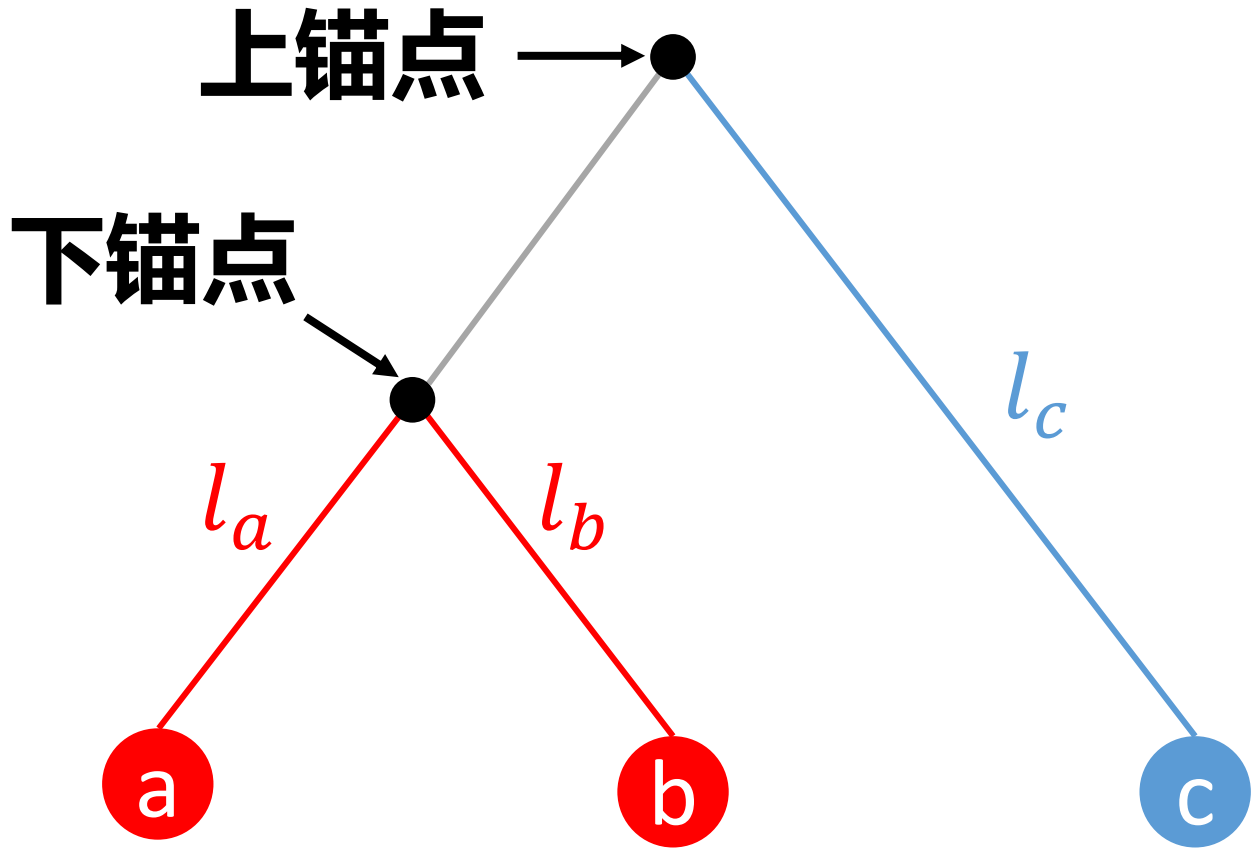
$$w_{\blacksquare} = 1_{uw} u_{\blacksquare} + 1_{vw} v_{\blacksquare}$$
$$w_{\blacksquare} = 1_{uw} u_{\blacksquare} + 1_{vw} v_{\blacksquare}$$
$$w_{\blacksquare\blacksquare} = u_{\blacksquare\blacksquare} + v_{\blacksquare\blacksquare} + 1_{uw} 1_{vw} u_{\blacksquare} v_{\blacksquare}$$
$$1_{vw} v_{\blacksquare} u_{\blacksquare\blacksquare} + 1_{uw} u_{\blacksquare} v_{\blacksquare\blacksquare}$$





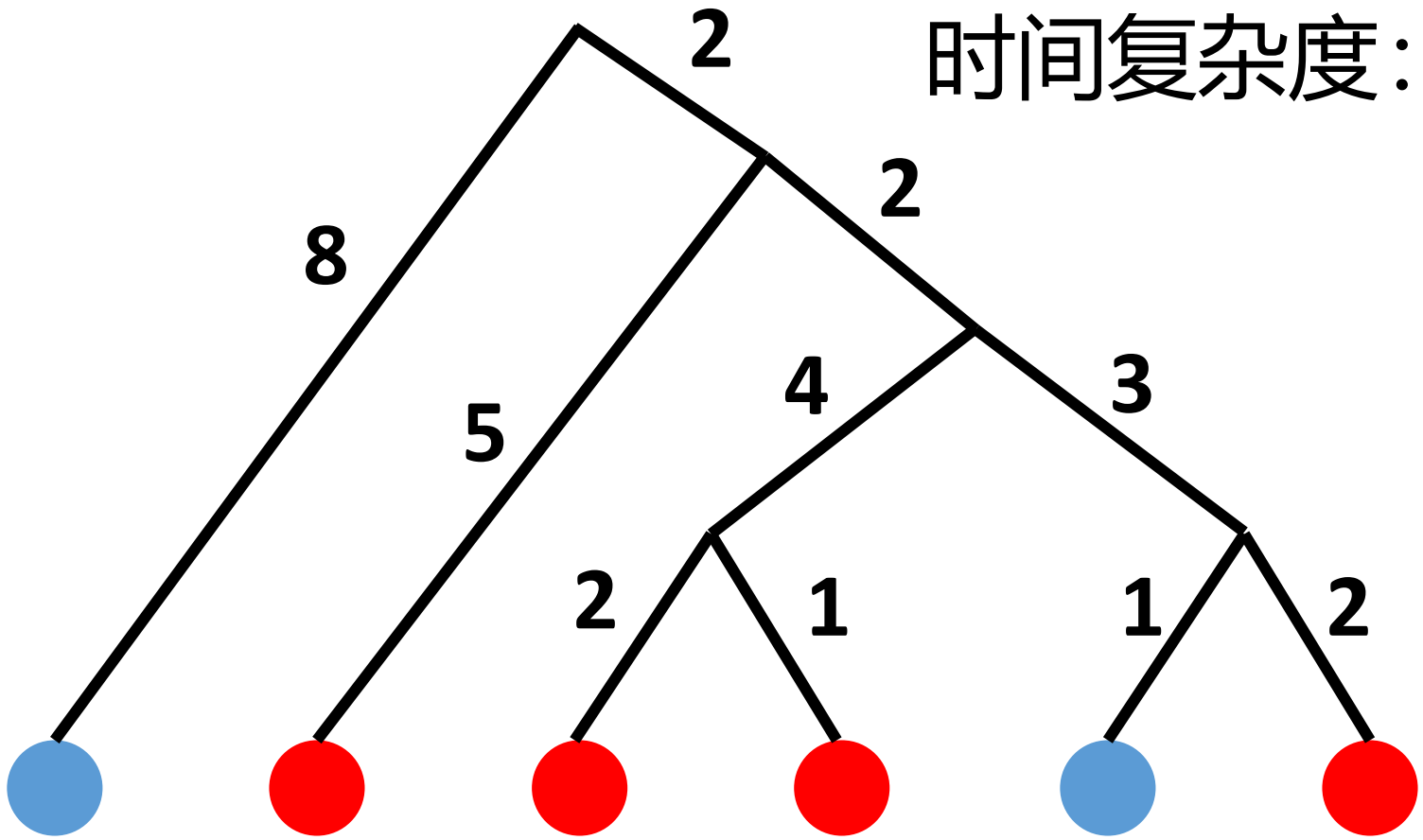
28.双色三叶枝长加权树问题

树的末端枝长与权重



左图所示的**三叶树**的**权重**为 $e^{-(l_a+l_b+l_c)}$。

27.1 红红蓝三叶树加权数量



时间复杂度：$O(n)$

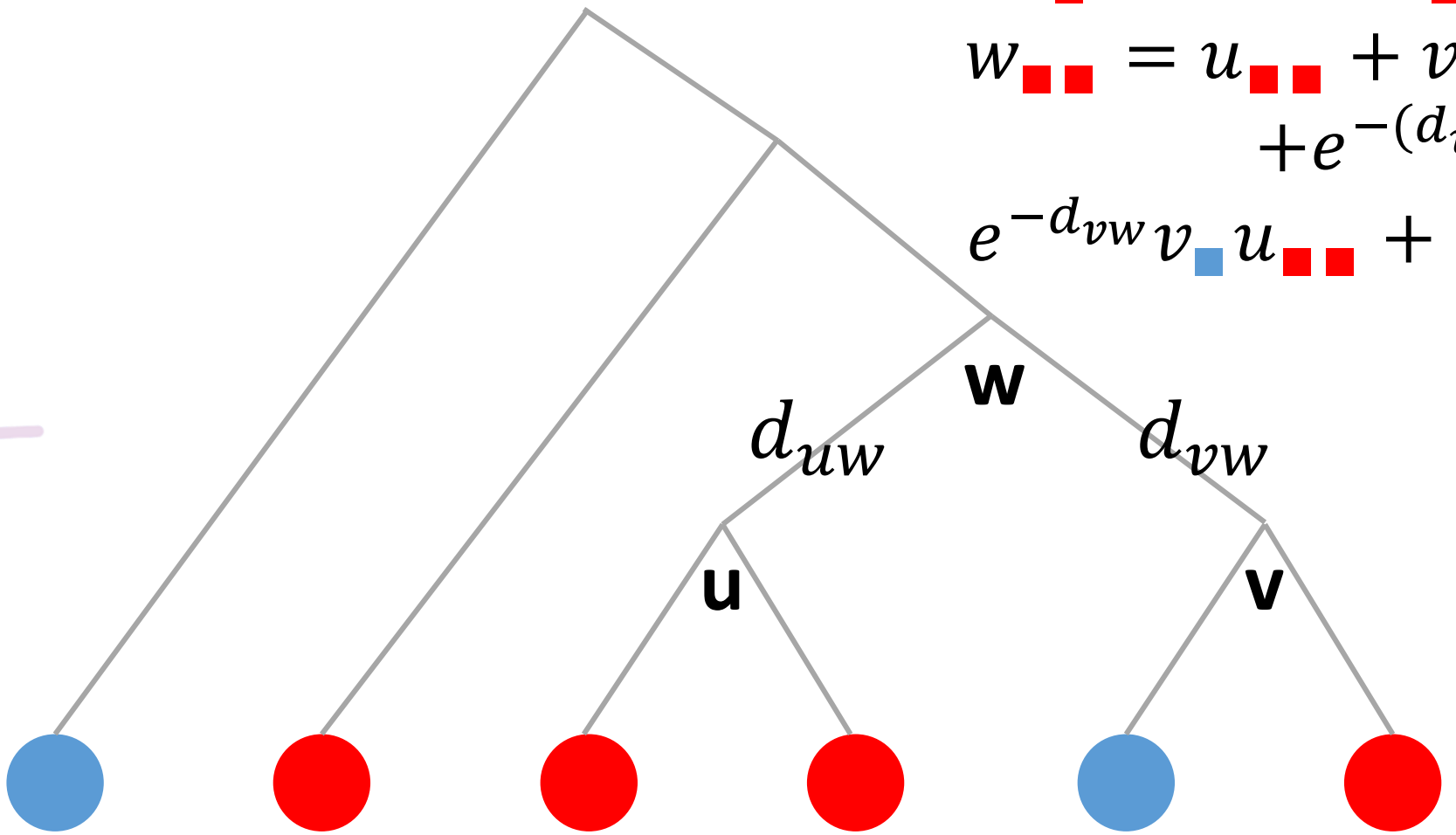
红红蓝三叶树算法

$$w_{\blacksquare} = e^{-d_{uw}} u_{\blacksquare} + e^{-d_{uw}} v_{\blacksquare}$$

$$w_{\blacksquare} = e^{-d_{uw}} u_{\blacksquare} + e^{-d_{vw}} v_{\blacksquare}$$

$$w_{\blacksquare\blacksquare} = u_{\blacksquare\blacksquare} + v_{\blacksquare\blacksquare} + e^{-(d_{uw}+d_{vw})} u_{\blacksquare} v_{\blacksquare}$$

$$e^{-d_{vw}} v_{\blacksquare} u_{\blacksquare\blacksquare} + e^{-d_{uw}} u_{\blacksquare} v_{\blacksquare\blacksquare}$$



北京大学
PEKING UNIVERSITY



上机教学

课堂练习

- http://aster.openjudge.cn/
- 本周按时完成课堂练习算作完成本周课堂互动。

三叶树叶节点放置（入门）

给定k棵有n个叶子节点的有根二叉树g1,…,gk，其叶子节点的标记集相同；给定一颗有n-1个叶子节点的有根二叉树s，其叶子节点的标记集和g1,…,gk比缺少叶子节点标记t。将标记t置于s的某处得到s’，使得s’与g1,…,gk的公共三叶树数量之和最大化。

课后作业

- 上机作业14-18

启发式算法评分赛

题目：可靠四叶最优树

给定**非二叉树**g_1，…，g_n，其中每条内部边被标记为实边（S）或虚边（D）。

寻找**二叉树**S，使得S与g_1，…，g_n的**公共可靠四叶树**数量之和最大化。

启发式算法评分赛

- 单人或两人组队，在微信群接龙获得队伍编号。
- 助教将队伍编号对应的基因树（输入）私发给每支队伍。
- 上机教学期间，每支队伍可将物种树（输出）私发给助教（请勿课下骚扰助教），以最后一次提交为准。
- 在三个课程节点，助教会公布每支队伍物种树与每个基因树的公共可靠四叶树之和，即该节点每支队伍的评分。
- 启发式算法评分赛的成绩根据三个课程节点的评分排名综合决定。
- 第一个节点：11月11日